

A Categorical Grammar View of Syntactic Categories: Evidence from Unexpected Coordinations

Pauline Jacobson
Program in Linguistics, Brown University
pauline_jacobson@brown.edu

1. Introduction

The goal of this paper is to show that under a rather natural view of syntactic categories compatible with the general theory of Categorical Grammar (hereafter, CG), some surprising and hitherto undiscussed (to my knowledge) facts about coordination fall out. It is standard in CG to maintain that there is a small set of primitive categories and a recursive definition of others, the latter are often referred to as 'function' categories (not to be confused with the notion of 'functional categories' in other theories). While expressions of one of these function categories have meanings which are functions, the use of the term 'function' in the syntax is sometimes metaphorical, or sometimes meant as functions from categories to categories (e.g., Steedman 2023). But here we advocate a view in which each such category corresponds to a function on strings: each function takes two strings (one at a time) and returns a third. We then show that combining this with a view making use of additional operations such as function composition (see, e.g. Steedman 1987, Dowty 1988) we make some striking predictions about what compositions are allowed and what aren't - predictions which are borne out by some subtle empirical facts surrounding coordination. A secondary result concerns a natural view of features on category labels in CG. By encoding them on the result categories of complex (function) categories, we successfully account for some unexpected good instances of coordination.

To give a spoiler alert: the view advocated here effortlessly handles the existence of surprising sentences like (1):

1.
 - a. Barry plays the dulcimer, the standup bass, and (he) does magic tricks.
 - b. Sarah studies syntax in the morning, semantics in the afternoon, and (she) practices the violin in the evening .
 - c. Barry plays the dulcimer, the bass, does magic tricks, and he builds treehouses.

Sentences like (1b) were first noted in Maxwell and Manning 1996 (hereafter M&M) but have received little attention,¹ and (1a) is a simplified version making the same point. It is surprising to see an ordinary object NP² coordinated with a full VP or a full S; (1b) has the extra fanciness of including 'nonconstituent coordination'. And note that the last S doesn't need to have the same subject; imagine (2) in the context of discussing a performing duo:

2. Barry plays dulcimer, standup bass, does magic tricks, and Patty plays crumhorn.

It should be stressed that these sentences (and others in this paper) crucially use what we can call 'comma' or 'silent' *and* for all but the last one. When we have a series of these I will refer to them as comma chains. Note that we also get comma *or* chains (with the *or* semantics throughout) as in (3); and (4) shows the M&M phenomena with *or* chains:

3. Lee, Sandy, or Kim will win the graduation prize.
4.
 - a. At the concert tomorrow if we want to spiff things up, Barry can play the sticks, the tambourine, or do magic tricks. (*NP, NP, or VP*)
 - b. At the concert tomorrow to spiff things up, Barry can play the sticks, the tambourine, do magic tricks, or Heidi can put on a silly hat. (*NP, NP, VP, or S*)

The reason I crucially use comma chains is that if we take (1a), for example, and replace the 'silent' conjuncts with overt *and*, the analysis would be quite mundane:

5. Barry plays the hammered dulcimer and the bass and (he) does magic tricks.

(5) simply involves coordination of NPs (actually, generalized quantifiers (GQs), see fn. 2), which means we have the complex NP (or, GQ) *the hammered dulcimer and the bass* which is object of *plays*. That VP in turn coordinates with *does magic tricks*. (In the case where the last conjunct is *he does magic tricks* we have S coordination.) If this were also the correct analysis of (1a) there would be nothing remarkable about (1). But one cannot simply assume that the cases above reduce to similar ones with overt *and* and *or*. This is because the silent versions have additional constraints on their distribution and meaning. For example, if we assumed that *the hammered dulcimer, the standup bass* was just ordinary NP (GQ) conjunction - exactly parallel to *the hammered dulcimer and the standup bass* - its well-formedness and interpretation would be independent of the fact this expression is followed by *and (he) does magic tricks*. We would thus incorrectly predict **Barry plays the standup bass, the hammered dulcimer* to be perfect, as well as **Tom, Dick left*. While some speakers do marginally accept comma chains with no overt *and*, these require more than two expressions in the chain as in *?Tom, Dick, Harry left*. (These also have a prosody suggesting an incomplete list.). More tellingly, even for speakers who do accept these, no one interprets them as having the meaning "Tom or Dick or Harry left", but nothing would preclude that if 'silent' *or* were just like overt *or*. And (1a) has no interpretation where the interpretation of the conjuncts is mixed; it cannot for example be understood as *Barry plays the dulcimer and the standup bass or does magic tricks* or as

Barry plays the dulcimer or the standup bass and does magic tricks. The empirical generalization is that a comma chain requires the presence of an overt *and* or *or* as the rightmost part, and the interpretation of the whole is governed by which is the overt conjunction. Thus (1) - (4) do not reduce to ordinary coordination of likes.³

There is a further surprise not noticed by M&M. This is the fact that when there is this type of comma chain, the 'bigger' categories can only be on the right. Notice that in all the above cases the constituents 'grow' as we move right. For example, (1c) and (2) are of the form NP, NP, VP, and S. But we cannot mix these to give the order VP, VP and NP with the obvious interpretation of his playing the standup bass:

6. *Barry plays the dulcimer, does magic tricks, and the standup bass.

Incidentally, (6) does have a pragmatically infelicitous interpretation where the last conjunct is interpreted as saying that he "does the standup bass". But as shown in Sec. 5.3, this (and other similar examples such as (49b)) is not a counterexample to the Grow Only Rightward Phenomenon as it has a different structure. In fact, the existence of that structure is itself (at first blush) another surprise. But the structure and reading also follows from how features can naturally be encoded on CG function categories, see Sec. 2.2. Thus the centerpieces of this paper are three main results. (a) The existence of M&M cases is unsurprising, and follows automatically from adopting a CG with function composition (or, it's "Curry'ed"equivalent). (b) The view of categories by which "function categories" correspond to functions on strings limits the types of composition which in turn accounts for the Grow Only Rightward Phenomenon. (c) A natural view of features in CG also accounts for some surprising good cases. And (a) and (c) receive independent confirmation

from the distribution of *either*. In addition to these results, we show (Sec. 4) an interesting fact about Right Node Raising (RNR) which also follow from this view of categories.

By way of a roadmap: Sec. 2 has three goals. For readers not familiar with CG, it lays out some basic background. Second, it crucially develops the main hypothesis in this paper: that function categories correspond to actual functions on strings. Third, Sec. 2.9 develops (without full detail) an account of leftward *wh*-extraction types of cases which differs from that in some other CG accounts (e.g., Steedman 1987 and many subsequent works) and is closer to the account in Gazdar et al. 1985. Sec. 3 goes on to discuss the status of the Coordinate Structure Constraint (CSC) - arguing with many others (e.g., Lakoff 1986, Kehler 1996) that the effects are not due to syntax. That section also develops an account of comma chains. Sec. 4 centers on the interaction of comma chains with RNR, showing that while there can be chains of conjuncts some with gaps and some without in RNR (akin to facts noticed in Lakoff, 1986 for "*wh*-extraction" cases) the rightmost one has to have a gap. This follows from the ingredients advocated here. The RNR case also contrasts interestingly with cases of "*wh*-extraction. The main punchlines are in Sec. 5, and center on what is outlined above in (a)-(c), and Sec. 6 shows that the distribution of *either* provides independent confirmation. The three appendices discuss the possibility of adding infixation operations in CG (Bach, 1979) and how that would interact with the constraints on function composition (Appendix 1); the hypothesis (argued against) that comma chains have a flat analysis (Appendix 2) and the hypothesis (also argued against) that M&M cases involve ellipsis and not function composition (Appendix 3).

2. A CG view of syntactic categories: functions on strings

For those not familiar with CG, this section introduces some of its fundamentals. At the same time, it introduces the view of categories advocated here (which is not one standardly adopted within CG, although it flows very naturally from the basic CG premises). Readers familiar with CG - especially as outlined in Jacobson 2014 - may wish to skim or skip some of the sections. To that end, I have marked those sections with the prefix (R), and have marked material that is new here or not well known with a prefix *.

(R) 2.1. Some basics of Categorical Grammar

We begin with some basics of the syntax (and semantics) of Categorical Grammar (CG). While there actually are many varieties of CG and related work in the Type Logical tradition, I base most of this on a version elucidated in Jacobson 2014. Our point of departure is hardly unique to CG: we take any linguistic expression (of any size) to be a triple of <[sound] (represented here with orthography), Category, meaning>. By meaning we mean a model-theoretic object – not a representation in a symbolic language, although such representations are used as ways to *name* model theoretic objects. One might wonder wherein is encoded the structure of an expression - i.e., its tree representation. Under the view here a tree is nothing more than a representation of how an expression is put together and plays no role in the grammar itself. Put another way: the grammar does not 'see' trees. (With one caveat: below and in Appendix 1 we explore the possibility that a small bit of internal structure needs to be kept track of for infixation operations.) Second, we can think of syntactic categories as encoding an expression's distribution. And in most versions of CG the category also encodes the type of meaning that an expression has; we return below.

Thus, there are a few primitive categories, and a recursive definition of all others. For the primitives, we begin with the set {NP (or, DP - see fn. 2), S, N, PP, CP}; there could be a few others. Other categories are defined recursively. As a first pass we say that if A is a category and B is a category then A/B is a category; call this a 'function' category. The interpretation of this is that an expression of category A/B wants to combine with an expression of category B to give as a result an expression of category A. (So, in the usual tree terms, the B-expression will be a sister of the A/B expression, and the mother will be A.) For example, we can think of a "VP" as being actually S/NP (or, S/X if categories besides NP occur as subjects), although I often use "VP" as abbreviation for this category.

***2.2. Features on Categories**

As is standard, we also assume that there is a set of features that can occur on these categories. Because the function categories are complex, an interesting question arises. Should the features always be encoded on (or as a part of) some basic category, or can a feature 'span' a function category such as A/B? It turns out that there are both empirical and theoretical reasons to suppose that in those cases where intuitively a feature 'spans' some function category, it is actually encoded on the result. By the 'result' category, we mean that basic category that we would get when an expression with a complex category takes all of its arguments. For example, take the category ((S/(S/NP))/N) (arguably the category of *every*). Here the leftmost S is the 'result'. To make the idea explicit, suppose there is a feature [INF] (for Infinitive) on VPs. The category of the VP then would actually be S[INF]/NP. Of course there can also be features on the argument category. For example S/NP[PL] would be the plural form of a verb, which means that it selects for a plural NP.

Even leaving aside the empirical benefits shown later, there are two reasons for this view. The first one follows from the logic of features in general. If even the so-called 'basic' categories like NP, S, etc. are actually themselves features (or feature bundles) - a view put forth originally in Chomsky 1970 - then having features always be a part of a basic category is exactly what we would expect. The extra features are really no different than the ordinary basic category features (again see Chomsky, 1970, Gazdar et al, 1985 for explicit discussion). That said, wherever we aren't spelling out a full category but instead using some label "X" as an abbreviation of a complex category, we write $X[F]$ for convenience, with the understanding that the feature F is actually on the result category. Also for expository convenience we will treat the basic category labels (e.g. NP) differently from the additional features. This is fairly customary. The second advantage of this view has to do with 'feature passing'. Consider a complex category like, for example, $(S/NP)/NP$ (the category for transitive verbs). Suppose the verb has some morphological form - again, say infinitive, where an infinitive S might be selected by something else. If the verb is $(S[INF]/NP)/NP$ there is nothing more to say about how the feature 'passes' from the transitive verb ultimately to the S. The featural information always passes up because it is always a part of the ultimate result category.

(R) 2.3. More on the function categories, and the syntax/semantics correspondence

Returning to more detail about the CG categories, note that merely saying something is of category A/B does not specify how this expression combines with a B to give A. We therefore elaborate the recursive definition of the 'function' categories as follows: If A is a category and B is a category then $A/_RB$ is a category and $A/_LB$ is a

category. The intuition behind these categories is that each encodes how the relevant expression combines with its 'argument'. An expression of category $A/_RB$ takes a B to its right to give as result an expression of category A , and an expression of category $A/_LB$ takes its argument to the left. (Other authors use other notations.)

This is all combined with a semantics. As noted above, the syntactic category of an expression also encodes its semantic type. We hypothesize that all expressions of a single category have the same type of meaning. Beginning with the basic categories, assume that any expression of some category A has as its extension a member of some set - call that a . All NPs, for example, have as their extension some individual; call the set of individuals e . Their full meaning is a function from worlds and times to individuals but we ignore intensions throughout - this again is just for expository convenience. Note that this means that a generalized quantifier like *every duck* is not an NP see Sec. 2.8. In addition, we take any category of the form A/B to have as its meaning some function from the set of things denoted by B -type expressions (call that set b) to some member of the set denoted by A -type things (call that a). Using standard notation, we say that an expression of category A/B denotes a function 'of type' $\langle b, a \rangle$ - meaning a function in $b \times a$. Thus assuming that the extension of any S is a truth value (a member of the 2-member set t), any S/NP (a VP) denotes some $\langle e, t \rangle$ (a function from individuals to truth values). This holds for both single word expressions like *walked* and more complex ones like *chewed gum*.

Thus *walks* has as its category $S/_LNP$ and a meaning of type $\langle e, t \rangle$. A transitive verb *likes* is of category $(S/_LNP)/_RNP$ and thus has as its meaning some function in $\langle e, \langle e, t \rangle \rangle$. Given that the meaning of any A/B is of type $\langle b, a \rangle$ and the meaning of its sister B is of

type b , it is quite unsurprising that the way the semantics works is that the meaning of the A/B expression takes the meaning of the B expression as argument. To make this explicit, we temporarily posit two rule schemas:

- (7) a. Given an expression a of the form $\langle [\alpha]; A/_R B; [[\alpha]] \rangle$ and an expression b of the form $\langle [\beta], B, [[\beta]] \rangle$, there is an expression of the form $\langle [ab], A, [[\alpha]]([[\beta]]) \rangle$.
- b. Given an expression a of the form $\langle [\alpha], A/_L B, [[\alpha]] \rangle$ and an expression b of the form $\langle [\beta], B, [[\beta]] \rangle$, there is an expression of the form $\langle [ba], A, [[\alpha]]([[\beta]]) \rangle$.

***2.4 Function categories as corresponding to actual functions on strings**

The two schemas in (7) look surprisingly alike – the only difference is in the directionality of the two expressions. This suggests that they should be collapsed. Of course, if we are just talking about having one collapsed rule vs. two rules, the difference is negligible. But if there are additional operations such as the case of a function taking its argument as an infix (Bach 1979) or the reverse - where the function itself is an infix - then there would be more rules to collapse. And the crucial result in this paper is that the tool needed to collapse these is exactly what leads to the predictions featured in Secs. 4 and 5.

Hence, we posit that what have been called "function categories" really do correspond not just to functions in the semantics (or functions from categories to categories) but - in parallel fashion - to actual functions on strings.⁴ The intuition here is that when the expression *walks* (a triple) combines with *Lee* (another triple) we have not only the result that in the semantics $[[walks]]$ applies to some individual l , but we also will have a function that maps the string $[lee]$ to the string $[lee walks]$.

A few steps are needed. First, we don't want the categories themselves to actually be functions; rather each function category corresponds to a function. This is because the basic categories obviously cannot be functions - in fact we can think of each such category as the name of a set of strings. This then should hold equally for 'function' categories; S/LNP names a set of strings (e.g., {[walks], [chews gum], [grunts],}). But it also corresponds to a function. Take the function corresponding to the category S/LNP . While one might think at first that that should be a function mapping NP strings to S strings, we need one more step. This is because the function corresponding to the category S/LNP can map [lee] to [lee walks] only if it first has access to [walks]. This is perfectly simple: a function corresponding to any category A/B is a function in $\langle \text{string}, \langle \text{string}, \text{string} \rangle \rangle$; it takes two strings as arguments (one at a time) to return a third. The first is the string category A/B , and the second is the string of category B, yielding a string of category A. By way of example, the function corresponding to S/LNP applies first to the string [walks] to give the function mapping [lee] to [lee walks], [sally] to [sally walks], etc. Some notation will be useful. For any (function) category X, we refer to the corresponding function as $F:X$. (read that as 'the function corresponding to the category X'). Similarly, for every function F which corresponds to a category, we refer to the category as $CAT:F$. We then define the relevant functions as follows:

$$(8) \text{ a. } F:A/_RB([x])([y]) = [xy] \qquad \text{b. } F:A/_LB([x])([y]) = [yx]$$

With that, we collapse the two rules in (7) as follows:

- (9) Given an expression a of the form $\langle [\alpha], A/B, [[\alpha]] \rangle$ and an expression b of the form $\langle [\beta], B, [[\beta]] \rangle$, there is an expression of the form $\langle [F:CAT_a ([a])([b])], A, [[\alpha]]([[\beta]]) \rangle$. (CAT_a means the category of a .)

To illustrate, *walks* is a triple of the form $\langle [walks], S/_LNP, [[walks]] \rangle$. (9) says this combines with a triple like $\langle [lee], NP, l \rangle$ to give $\langle [lee walks], S, [[walks]](l) \rangle$.

(R) 2.5. Word order rules

The ultimate word order in a sentence is determined by the categories of the lexical items. But obviously these should not be listed on a case-by-case basis. For example, we clearly don't want every intransitive verb in English to just happen to be listed in the lexicon with category $S/_LNP$ as opposed to $S/_RNP$. To this end, Jacobson 2014 suggests that lexical items are (in general) 'born' in an underspecified form and thus without directional slashes, and those are added by general rules. A first pass at these rules would be as follows (from Jacobson 2014); the on either side of the category means that these rules apply even when that category is embedded in others. The intent also is that the directional slashes are only added where there are none; no rule overrides already assigned directional features:

- (10) a. $\dots(S/X)\dots \rightarrow \dots(S/_LX)\dots$ b. $\dots(X/X)\dots \rightarrow \dots(X/_LX)\dots$
 c. The default rule: $\dots(A/B)\dots \rightarrow A/_RB$.

Being the default, the rule in (c) has to apply after (a) and (b). This list is incomplete but will serve for now (see Jacobson 2014 for prenominal adjectives).

*2.6. *and* and *or*

We turn next to the category for *and* (and *or*). We assume that *and* is listed in the lexicon as a cross-categorical item: $(X/X)/X$, with the generalized (Curried) semantics

give would have the category $((S/LNP)/_I NP)/_R PP$ (borrowing familiar terms from morphology, this means that *give to Fido* is a circumfix).

To fully work out the details of a system with infixation, we also need to assume that expressions are not just strings, but strings with a dedicated infixation point or, as in Pollard 1984, strings with additional information from which can be derived an infixation point. (Hence when *give to Fido* combines with *the bone*, the grammar needs to know that the latter is infixed after the verb). Thus Pollard 1984 takes each string to be a headed string, and derives the infixation point from that. Assume that infixes in English go to the right of the head. As example, *give to Fido* is a string headed by *give*, so *the bone* is infixed in to the right of *give*. Also needed are conventions as to whose infixation point - or head - is inherited in the new expression. I assume in general this is inherited from the function category (but see the remarks below on Lift). And a full system with infixation also needs a 'circumfix slash' $(A/_c B)$ indicating that the expression with this category is itself an infix - i.e., it takes its argument as a circumfix. There remain pieces to be fully incorporated into the view of categories in this paper, but a complete treatment of a grammar with infixation is beyond the present scope (see Jacobson, Li and Robertson in preparation). Since infixation rules do potentially interact with some of the points here we will note those as relevant; to keep the flow manageable, we leave those issues largely to Appendix 1.

2.8. Additional combinatory rules and applications to conjunction

(R) 2.8.1. Lift

Consider the syntactic category of generalized quantifiers like *every dog*. Under the view of the syntax/semantics match here, these cannot be NPs as they do not denote individuals.

Using the fairly standard wisdom since Montague 1973, their semantic type is $\langle\langle e, t \rangle, t \rangle$. In subject position they take the VP as argument which leads us to the conclusion that their syntactic category is $S/R(S/LNP)$, and a quantificational determiner like *every* is of category $(S/R(S/LNP))/RN$. (This says nothing about generalized quantifiers in object position; see Hendriks 1993 and Jacobson 2014 for relevant discussion). But note an anomaly about this category: the directional slashes are not expected. The L-slash is not surprising and follows from the word order rules given in (7), and the directional feature on the final slash in *every* is the unsurprising default R feature. What is surprising is the first (reading left to right) R slash. Should this not be an L-slash, since in general whatever combines with any X to give an S takes that S to the left? We return to this directly.

Partee and Rooth 1983 proposed that ordinary NP meanings (individuals) can also lift to generalized quantifier meanings (type $\langle\langle e, t \rangle, t \rangle$) (i.e., the meaning in Montague 1973) such that if the name *Lee* denotes the individual *l*, then the lifted meaning of *Lee* is $\lambda P[P(l)]$ – i.e. function mapping any $\langle e, t \rangle$ function *P* to the value that *P* assigns to *l* (in set terms, the set of sets with *l* as member). This was motivated by the fact that *Lee*, for example, can conjoin with a generalized quantifier (*Lee and every candidate for schoolboard will speak at the meeting*). But rather than simply list the meaning of *Lee* in the lexicon as the 'fancier' meaning (as Montague did), Partee and Rooth argued that it is 'born' with the simpler meaning of type *e*, and can lift to the $\langle\langle e, t \rangle, t \rangle$ meaning. We adopt this proposal, and given the rest of the premises of CG, the lift rule must also map the syntactic category NP to the generalized quantifier category $S/R(S/LNP)$ (call that GQ).

Moreover, we assume that the rule is much more general (for some motivation see Dowty, 1988), hence we allow any expression to lift. By way of formalizing this, notice that lift is an operation which is defined perfectly generally and is independent of its use in grammar. Take any set A , any member a of this set, and any set of functions $\langle A, B \rangle$ (i.e., the set of functions in $A \times B$). Then we define $\text{Lift}(a)$ as that function taking as argument any function f in $\langle A, B \rangle$ and mapping f to the value that f assigns to a . The result is thus something in B , so $\text{Lift}(a)$ is a function in $\langle \langle A, B \rangle, B \rangle$.⁷ Restating this informally, we start with some object $\underline{a}$. The lifted $\underline{a}$ is now the ‘boss’, it takes some function f (that could have taken $\underline{a}$ as argument), and gives back just what would have resulted had f applied $\underline{a}$.

Given the rest of this version of CG, we also want to tie this in to the syntax. Hence we can write a rule mapping a single triple into a new one, differing in meaning and category but not in sound. This is an instance of what is often called a type shift rule, but we avoid that terminology as the rule it also affects the syntactic category. Note further that there can be no notion of ‘type shift as a last resort’ in the architecture of the grammar assumed here. That makes sense only if the syntax precomputes representations which are ‘sent’ to the semantics whose job is to compositionally interpret these (so if it encounters a type mismatch, it has at its disposal a set of type shift rules to ‘repair’ this). Here the syntax and semantics are computed in tandem so the notion of ‘repair’ makes no sense. The general lift rule is (preliminarily) given in (12). Note that because there are two possible outputs in terms of the syntactic category, this is actually two rules which we exhibit together:

- (12) Given a linguistic expression of the form $\langle [a], A, [[a]] \rangle$, there is an expression b of the form $\langle [a], B/L(B/R A) \text{ or } B/R(B/L A), [[\text{lift}_{\langle a, b \rangle}(a)]] \rangle$.

*2.8.2. Lift as order preserving follows from the view of categories

The only difference between the two rules in (12) is the syntax of the new category. Aside from the discomfort of having two lift rules packed in here, this looks suspicious for another reason: why in both cases does the new category just happen to preserve word order? For example, when *Lee* lifts over VPs, why is the result of combining lifted *Lee* with *walks* exactly the same as *walks* combines with unlifted *Lee*? Is it just a stipulation that there is no way to apply lift to yield, for example, $B/L(B/LA)$ as the new category? The answer follows from the view of categories advocated here. If we take seriously the notion of syntactic categories as corresponding to functions, then the word order preservation property of Lift is no accident. By definition, a lifted object takes some function as argument to give exactly what would have resulted had that function taken the original object as argument. Hence we merely need to extend this notion to the syntactic categories. The translation into categories is, unfortunately, not immediate for two reasons. First the categories themselves are not functions but correspond to functions, Second, there is the extra layer that the function first needs to apply to the string (sound) part of the expression in order to be a function of the right type. But we can nonetheless extend the basic idea of lift to syntactic categories as follows. First we define $F'-\alpha$ as that function in $\langle \text{string}, \text{string} \rangle$ obtained by applying $F:CAT_\alpha([\alpha])$. In prose, $F'-\alpha$ is that function from strings to strings that results from applying the function corresponding to the category of α to the string $[\alpha]$. Since the syntactic part of the lift rule maps a category A to something that takes expressions of category B/A as argument, we notate that as $Lift_{A/B}$. With this we can see the sense in which the syntactic lift is the same as the general lift operation:

- (13) Given any category A, $\text{Lift}_{A/B}(A)$ is that category C such that for all expressions α such that $[\alpha]$ is in A and all β such that $[\beta]$ is A/B, $F:C([\alpha])([\beta]) = F'-\beta([\alpha])$.

(Recall that the categories are also names of sets of strings.) In other words, the new category C is such that for any string $[x]$ in A and any string $[y]$ in A/B, applying the function corresponding to C to $[x]$ is a function mapping $[y]$ to the string which is the result of applying $F:A/B$ to $[y]$ and then to $[x]$. As an example, since *F-walks* maps $[\text{Lee}]$ to $[\text{Lee walks}]$, then the lifted category (CAT) of *Lee* is such that $F:\text{CAT}$ applied to $[\text{Lee}]$ maps $[\text{walks}]$ to $[\text{Lee walks}]$. From this, it follows that the lifted category can take an expression of category B/A to its right only if that expression had a category looking for an A to its left, and vice versa. It is thus not a 'bug' that lift happens to preserve order; it follows from the general definition of lift extended to categories. Note that if we have infixation slashes and corresponding circumfix slashes those would also fold into the system: A could lift to $B/_C(B/_IA)$ or to $B/_I(B/_CA)$.⁸

We now return to the earlier puzzle of why a quantificational determiner has the surprising category $(S/_R(S/_LNP))/_RN$. Recall that most of those directional features are predictable, but the first (in the left-to-right sequence) R-slash is surprising. Since the last argument in forming an S is usually taken to the left, why would the VP be taken to the right here? Similarly, in the case of a lexical generalized quantifier like *nobody* why is its category $S/_R(S/_LNP)$? The answer for the case of *nobody* is that this is simply listed with category $\text{Lift}_{S/NP}(NP)$. It follows from that that it can be $S/_R(S/_LNP)$. It could also be $S/_L(S/_RNP)$. Similarly, a quantificational determiner such as *every* is listed in the lexicon with category $\text{Lift}_{S/NP}(NP)/N$. Incidentally it doesn't matter that there are items listed as

being of category $\text{Lift}_{S/NP}(NP)$ which don't have identical strings of category NP (i.e., the generalized quantifiers), since the category $\text{Lift}_{S/NP}(NP)$ is still perfectly well defined.

(R) 2.8.3. Function Composition

Steedman 1987, 2023 and many works in between proposed adding function composition as a possible combinatory rule, and showed how this can account for leftward 'wh'- extraction cases. As will be detailed in Sec. 2.9 I actually treat leftward (wh-) extraction differently; one new rationale for this is given in Sec. 4, and greater detail can be found in Jacobson (to appear). Nonetheless, having function composition (or a variant thereof to be discussed below) has striking payoffs especially with respect to coordination.

The general operation of function composition is defined as follows. Given a function f in $\langle a, b \rangle$ and a function g of type $\langle b, c \rangle$, then $g \circ f$ (read “g compose f”) is a function in $\langle a, c \rangle$ and is $\lambda x_a[g(f(x))]$. Initially then we add two general rules to the combinatory apparatus:

- (14) a. Given an expression α of the form $\langle [\alpha], A/RB, [[\alpha]] \rangle$ and an expression β of the form $\langle [\beta], B/RC, [[\beta]] \rangle$, there is an expression of the form $\langle [\alpha\beta], A/RC, [[\beta]] \circ [[\alpha]] \rangle$. (“right composition”)
- b. Given an expression α of the form $\langle [\alpha], A/LB, [[\alpha]] \rangle$ and an expression β of the form $\langle [\beta], B/LC, [[\beta]] \rangle$, there is an expression of the form $\langle [\beta\alpha], A/LC, [[\beta]] \circ [[\alpha]] \rangle$. (“left composition”)

With the addition of these two rule schemas, a sentence like *every candidate tells lies* has two possible derivations giving the same word order and the same meaning. The familiar derivation is the one in which *tell* (of category $(S/LNP)/RNP$) first combines with

the NP *lies*. This yields *tells lies* of category S/LNP which is taken as argument of *every candidate* of category $S/R(S/LNP)$ with the familiar semantics. Alternatively, *every candidate* can function compose with *tell* as the reader can verify to give the S/RNP *every candidate tells*. When this takes *lies* as argument, we get the same meaning as the 'traditional' derivation. Indeed this follows by the very definition of function composition, hence $[[\text{every candidate}]] \circ [[\text{tells}]] ([[lies]]) = [[\text{every candidate}]] ([[tells]] ([[lies]]))$.

The same possibilities exist with an ordinary subject like *Lee*. Here the minimal derivation is the ordinary one where *tells* combines with *lies*, and *tells lies* takes as argument the NP *Lee*. But *Lee* can also lift to the generalized quantifier category, in which case *Lee tells lies* has the two derivations exactly analogous to the case of *every candidate tells lies*.⁹ In particular there is an analysis by which *Lee tells* is a well-formed expression.¹⁰ One benefit of these 'extra' and unusual constituent structure analyses is that the existence so called non-constituent coordination and/or Right Node Raising follow effortlessly. Thus a typical RNR example like (15) is simply the coordination of two expressions of category S/RNP which then ultimately takes the NP *model theoretic semantics* as argument:¹¹

(15) Lee loves and Sandy hates model theoretic semantics.

In addition, Dowty 1988 showed how lift and function composition together allow for a constituent like *lobster on Tuesdays* in (16), which in turn predicts the existence of surprising so-called non-constituent conjunction cases like (15):

(16) Captain Jack serves scallops on Mondays and lobster on Tuesday.

on Monday is a VP modifier; i.e. VP/LVP . *scallops* is an NP, but can lift over transitive verbs to category $VP/L(VP/RNP)$ or - for short VP/LTV . These two combine by the left

composition rule in (13b) yielding *scallops on Monday* of category VP/LTV . Thus this (like lifted *scallops* itself) wants a transitive verb to its left to give a VP. *lobster on Tuesday* is formed similarly, the two conjoin, and take the transitive verb *serves* to the left.^{12, 13}

* 2.8.4. Function Composition as function composition

Let us return now to the two function composition schemas in (14). Restricting ourselves to categories making use only of R and L slashes (thus leaving aside the possibility of infixation) the main claims of this paper are that (a) these are the only two types of function composition and (b) that this follows from the view of syntactic categories advocated here. The bulk of Secs. 4 and 5 will justify the claim in (a). Focusing here on (b), we claim that when an expression of category A/RB composes with one of B/RC the result is A/RC and not A/LC , and that this is no accident. Moreover - unlike the view in Steedman 2023 and other works - we also claim that there is no composition such that an expression of category A/RB can take to its right an expression of category B/LC to give A/LC (and where the semantics is function composition). We refer to this as S-crossing composition. Steedman also allows the mirror image, whereby an A/LB can take to its left a B/RC to give an A/RC (also here called S-crossing composition). A major claim in this paper is that there is no S-crossing composition. Before we return to the question of why that should be so, we can see an immediate benefit to disallowing this under the assumptions here. Thus having S-crossing composition would allow *believe* (of category VP/RS) to take *lost* (category S/LNP) to its right to give *believe lost* of category VP/LNP . That then would give a well-formed VP *Joe believed lost* and thus ultimately sentences like **Lee Joe believed lost* (meaning "Lee believed that Joe lost"). Incidentally, this is not to

say that it is inconceivable that *believed lost* can combine - as is well know, this is found, for example, in leftward *wh*- type extraction cases (we return in Sec. 2.9). But while this may be a well-formed expression, we don't (under the system here) want this to have the category $VP/_LNP$ which is what would result if S-crossing composition was available.¹⁴

If we are correct that this type of "composition" is precluded, why is that? Put differently, why are the only instances of syntactic function composition (as demonstrated by coordination possibilities) just those that preserve word order? Once again, the answer lies in the view of syntactic categories assumed here. If categories correspond to actual functions in $\langle \text{string}, \langle \text{string}, \text{string} \rangle \rangle$ and 'function composition' is literally the composition of two functions (resulting after the function category is applied to the string in question), then the schemas in (13) are the only two possibilities using only R and L slashes. (Other types are allowed if there are C and I slashes; see Appendix 1.2.) The key point here is that S-Crossing composition is not allowed). With this, we reformulate (14). Recall that for any α of category X, $F'-(\alpha)$ is that function which results when $F:X$ is applied to $[\alpha]$, hence (14a and b) can be recast as (17):¹⁵

(17) Given an expression a of the form $\langle [\alpha], A/B, [[\alpha]] \rangle$ and an expression b of the form

$\langle [\beta], B/C, [[\beta]] \rangle$, there is an expression γ of the form $\langle F'-(\alpha) \circ F'-(\beta) ([\alpha])([\beta]), CAT$
of γ is such that $F'-(\gamma) = F'-(\alpha) \circ F'-(\beta), [[\alpha]] \circ [[\beta]] \rangle$.

Since Steedman 2023 and other works makes crucial use of S-crossing composition, it is incumbent on the theory here to show how the phenomena he accounts for have other analyses. Space obviously precludes exhaustive discussion but we mention one prominent case; Steedman uses 'second order' S-crossing composition to account for Dutch and Swiss

German cross-serial dependencies. But Pollard 1984 showed that these can also be handled using a Wrap operation, so these can be accommodated without S-crossing composition.¹⁶

***2.9. "Wh" (leftward) Extraction**

In the CG and related literature there has been debate as to whether the ordinary 'slash' of CG (indicating an argument slot) is to also be used for the kinds of "leftward extractions" that go under the rubric of *wh* extraction (by which I include Topicalization, even though it contains no overt *wh*). Gazdar et al 1985 and earlier Gazdar 1979 clearly hinted at the similarity by giving the GPSG extraction feature the name 'slash'. But those proposals were not embedded in a Categorical Grammar, and so the two were treated differently, and there was no claim that an argument slot 'missing' in a *wh*-type extraction construction had the same status as an expected argument in CG. However, Steedman 1987 famously united them within CG by simply having many items select (to their right) something with a right slash. For example, a question word like *who* can have as its category $Q/R(S/RNP)$ - it takes to its right an S/RNP to yield a question, where the expression of category S/RNP is composed in exactly the same way that we find in, e.g., RNR cases. This idea is certainly appealing: all other things being equal, the Steedman approach is obviously simpler than having the argument structure slash and the 'extraction slash' be different. That said, all other things are not equal. It has often been noted that there are differences between RNR (which in the framework here is a matter of conjoining two things with a right slash) and *wh* extraction (see, e.g., McCawley 1982, McCloskey 1986). One well known example is that extraction is possible in cases like *believe* ___ *was guilty*, but neither Heavy NP Shift nor RNR are allowed:¹⁷

- (18) a. Who do you believe ___ was guilty of cheating?
 b. *I believe was guilty of cheating the student who sat in the front row.
 c. *I believe was guilty of cheating and know should be reported to the dean the student who sat in the front row.

Note that we have already shown that *believe was guilty of cheating* cannot be put together under the view of function composition here to give a VP/LNP. By the same logic it cannot be put together to give VP/RNP, so the badness of (b) and (c) are expected. (That this cannot be a VP/RNP follows equally well in a system with S-crossing composition.)

Hence I follow Oehrle 1990 in separating these within a CG framework. I use the notation | to indicate an 'extraction' gap; one can think of this as akin to the superscript notation in Jacobson 1999 which contains the information that an expression contains within it a pronoun (unbound within it). For the purposes of this paper, we will not think of either | or Jacobson's pronoun superscript as actual features, although this probably would do no harm. Thus we expand the recursive definition of categories: for A and B both categories, there are categories of the form A/B (with R, L and possibly I and C on the 'slashes'), a category of the form A|B and a category of the form A^B. The semantic type of all of these is <b,a> but the syntax is different. An expression of category A/B expects to find an actual B to combine with. Of course it might not; it might instead function compose, or itself be argument of some other function. (In CG in general there is no analog to anything like a Projection Principle or Theta criterion - nothing requires functions - either in the syntax or the semantics - to be 'saturated'). An expression of category A^B

contains (usually) within it an unbound proform of category B (and again denotes some function in $\langle b, a \rangle$) and $A|B$ simply indicates an A-like expression but one 'missing' a B.

In terms of how the $|$ is 'percolated', we adopt here a fairly simple set of conventions. First, let any A/B to map to $A|B$; this is what Gazdar et al. 1985 called the 'bottom' of an unbounded dependency construction, and allows an expected argument to instead be encoded as a "gap" (or, "missing argument"). Assume additionally that for every A/B there is also a variant $(A|C)/(B|C)$. This is how the information that a daughter has an 'extraction' gap is passed to the mother. The mapping of A/B to $A|B$ has no effect on the semantics (since both are of semantic type $\langle b, a \rangle$), while the semantics of the second mapping above is the "Geach" rule (g , see fn. 15). This is a unary version of function composition, hence for any function h in $\langle b, a \rangle$, $g(h)$ is a function in $\langle \langle c, a \rangle, \langle c, b \rangle \rangle$ such that $g(h) = \lambda X_{\langle c, a \rangle} [\lambda C [h(X(C))]]$. All this is undoubtedly too general: it allows *wh*-extraction gaps virtually anywhere. It provides no account of typical island effects (but see fn. 17), and it does not build in a Left Branch condition. While the latter would be easy enough to do by restricting the $|$ 'passing' rule above to $(A|C)/_R(B|C)$, we do not do that here since the examples below regarding "Lakoff chains" show that gaps can indeed be on a left branch only. It also provides no account of *that-trace* effects, etc. But we leave those matters as this is not the main focus of the account, and for now content ourselves with what is surely an overly permissive account of *wh* extraction. One other point to note is that ATB extraction is - just as in Gazdar et al 1985 - unsurprising. It just involves the coordination of likes. The lexical specification of *and* is $(X/X)/X$, and X throughout can be instantiated by a category of the form $A|B$.

3. Coordinate Structure Constraint, Comma chains, and Lakoff chains

3.1. The coordinate structure constraint

While it has of course been known since Ross 1967 that extraction from just one conjunct often gives rise to bad outputs, there are two major strategies for accounting for this. One posits something extra in the syntax; the other assumes that these are not about the syntax at all but rather about information structure (or 'conceptual structure') (Lakoff 1986, Kehler 1996, Culicover and Jackendoff 1997). We will not attempt to propose or endorse any particular full information structure (or similar) account here, but will be providing new evidence that CSC effects should not come from a principle hardwired into the syntax given how easy it is (with the right construction) to come up with counterexamples. The strategy for constructing the kinds of counterexamples in this paper was first taken in Lakoff 1986 and involve violations of the CSC in comma chains. These provide a fertile ground for many extensions of his observations as will be shown below.

Before turning to these, it is worth pointing out why it is that CSC effects are not automatic in theories like G/HPSG and varieties of CG. One might think they are - given the "fact" that only like categories conjoin, combined with the idea that a category like $A|B$ or A/B is not the same as the category A . Indeed, that was an early hope in, e.g., Gazdar 1981, but had been abandoned already by Gazdar et al. 1985. The reason that these effects do not follow from the idea that 'only like categories conjoin' is that it is a mistake to think that 'only like categories conjoin' itself is hardwired into the grammar. In fact, without an extra principle, it is just (often) a consequence of the lexical specification of conjunctions (i.e., in CG $(X/X)/X$ and similar lexical entries in other theories), and/or the phrase

structure rule schemas (or modern equivalent) allowing any rules like $X \rightarrow X \text{ Conj } X$ (or two binary rules to do the same thing). But the lexical categories in CG and/or the phrase structure rules (or their modern equivalent) in, e.g., GPSG, HPSG, and movement theories are exactly what *in general* can be overridden by 'slash passing' conventions, or function composition, and/or movement. Indeed, that is the very rationale for these devices. A verb like *think* expects to take a full S to yield a VP, but of course its complement (on the surface) can be missing something, so we posit either a 'slash', function composition, movement, or some other such device. So why should *and* be any different? Thus, a principle to the effect 'only likes can conjoin' would have to be something extra - some sort of output filter. This was already recognized by Gazdar et al 1985, who instead accounted for CSC in a different way. In Steedman 2023 the CSC effect is accomplished by a new kind of slash which resists composition; this is used for both the right and left argument slashes of conjunctions. But that too is extra.¹⁸ So if we were to discover that the syntax does not have to be responsible for CSC effects where they occur, that should be a welcome result.

Thus there is a large body of literature spanning decades speculating that CSC effects are about information structure rather than syntax, along with a large body of counterexamples to claim that the CSC is hardwired into the syntax (regardless of the particular theory). A typical and often cited one is (19), (20) is along the lines of examples from Goldsmith 1985 (who defends the CSC):

(19) What did Lee go to the store and buy ___?

(20) How much beer can Lee drink ___ and still stay sober?

Though less discussed, one can also construct counterexamples with *or*:

(21) This is the car that Lee will (either) buy ___ or else he'll just keep taking the bus.

One reaction to these counterexamples is to posit a different *and* in these examples, and hence presumably also a different *or* in (21) (see Goldsmith 1985 for a restructuring analysis). For a case like (19) it is sometimes thought that this is a 'subordinating' *and* rather than a coordinating *and*; the hope would be to extend that to (20) and (21). But Culicover and Jackendoff 1997 convincingly argue against any account whereby *and* subordinates the left conjunct. And it is not clear what it would mean for there to be a different *and* which subordinates the right conjunct, since under many modern views of the structure of *and* - including the view here - that is the case for ordinary *and* (any theory with only binary rules has as a consequence that one conjunct is 'subordinated' to the other, and the normal word order rules of English predict that that is the right conjunct.). Note incidentally the CSC violations need not involve only VP conjunction; (22) - (24) are S conjunction even with different subjects, nor does there need to be a coreferential pronoun in the second conjunct as long as the connection between the two is clear:

(22) How many banks can Clyde rob ___ and the police still not arrest him?

(23) How many banks can Clyde rob ___ and Bonnie still pretend that all is fine?

(24) This is the castle that Charles will (either) decide to keep ___ or (else) William and Kate will just have to give up their dreams of living in a castle.

And what about the new *or* needed in (21) and (24)? It seems to have the same meaning as ordinary *or*. Again there is a lengthy literature on this (with respect to *and*) and I have nothing new to offer on just what is the needed information structure; see e.g. Lakoff 1986, Kehler 1996, Altshuler and Truswell, 2022 among many others.

3.2. Lakoff Chains

What I take to be especially suggestive evidence against the claim that there are different items *and* and *or* (crucially with a different structure) is that we find the same CSC-violating possibilities in comma chains, which involve null *and* and null *or*. This was pointed out in Lakoff 1986 (for the *and* case; he did not look at *or*). His own argument using these examples against a syntactic CSC was different than the point to be made here; I nonetheless refer to these as 'Lakoff chains'. Hence I define a *Lakoff chain* as any comma chain in which some of the conjuncts contain 'missing material' and some don't (including the case in Sec. 5 where some are missing material on the left which others are not). Thus for the case of *and*, Lakoff gives examples along the lines of those in (25) (I indicate which expressions have gaps and which don't by use of __, with no intended theoretical significance). (26) shows the same point with *or*:

- (25) a. How much/What kind of beer did Lee go to the store, buy __, pack up __,
and then bicycle home?
- b. How much beer will Lee buy __, drink__ and still not fall asleep?
- c. How much/What kind of beer did Lee buy __, bicycle home, and then instantly
proceed to drink __?
- (26) This is the car that Lee will (either) go ahead and buy __, decide he doesn't actually
have enough money to spend on silly things, or convince his father to give him __.

I put *either* in (26) to make it more natural and easy to process as it signals right away that the chain is an *or* chain, but that is not strictly speaking necessary.

Consider the implications of the claim that the above all involve a different *and* or *or* whose syntax is such that they escape CSC violations. Presumably they would also have different meanings from ordinary *and* and *or*, but it is difficult to imagine what each would mean. Notice that ordinary (Boolean) *[[and]]* and *[[or]]* are sufficient to get the right semantics for (25) and (26). Even more problematic is the observation earlier about comma chains: the last member must be overt and it determines the interpretation of all the rest of the chain (all *[[and]]* or all *[[or]]*). This holds equally well in the Lakoff variants; (26) for example is interpreted as three disjuncts. (For (26) one might argue that this is due to the pragmatics, but the same holds for *This is the car that Lee will rent ___for a month, buy ___, or decide to just learn to bicycle*. The first two alone can felicitously have either an 'and' interpretation or an 'or' interpretation, but the last disjunct forces 'or' throughout.) In the next section we develop a set of conventions to ensure this in comma chains. But under almost analysis of comma chains, if Lakoff chains involved different conjunctions with different syntax we would need a new set of conventions for their behavior.

Interestingly, it has often been proposed that there is an additional item pronounced *and* which is the 'sum' *and* (Link 1983). It takes two individuals (atoms or plurals) as input and returns a Linkian plural; since those are individuals further sum conjunction is possible. (Recall that the ordinary Boolean *and* never conjoins NPs because meet is not defined for individuals; cases of apparent NP conjunction actually have to be the conjunction of GQs. But sum *and* conjoins individuals.) Sum *and* also has a null variant, as we find comma chains like *Groucho, Harpo, Chico, and Zeppo made a great comedy team*. And, not surprisingly, if the rightmost conjunct has *and*, the chain has to have either the Boolean

and or the sum reading throughout. (See Borsely 2005 for related observations.) Thus consider a ride sharing coop where one accumulates hours by driving seniors to medical appointments, etc. Once one has accumulated 100 hours, one is entitled to the same services when needed. Suppose that the hours can be accumulated both by individuals or by teams, with the number of benefit hours being doled out to the individual or to the team. (Note that one might think of this as always teams, where the individual is just a one-person team. That will not affect the point.) Let Lee and Sandy be a team who together have accumulated 100 hours, and let Jessie be an individual (or singleton team) who also has. I can report (with the right prosody) *Lee and Sandy and Jessie have reached the goal*. But I cannot report this as *Lee, Sandy, and Jessie have reached the goal* where the overt *and* is Boolean *and* (hence both the 2-person team and the individual/singleton team met the goal) but where *Lee, Sandy* is sum *and*. The truth conditions here require each of Lee and Sandy and Jessie to have put in 100 hours, or that all three constitute a team.¹⁹

Strikingly, sum *and* also participates in Lakoff chains:

- (27) This is the student who the mother of, the father of, and the principal of North High became an ad hoc committee to investigate the students' violent drawings.

None of this is surprising. Other than the fact that sum *and* conjoins NPs and Boolean *and* conjoins other categories, their basic syntax appears to be the same. But if the CSC violations found in Lakoff chains were due to a different syntax then there would be three new conjunctions looking just like Boolean *and*, *or*, and sum *and* all three of which would have a new syntax allowing for CSC violations, and all of which would necessitate a new account of comma chains with these new items.²⁰ Henceforth we assume that the CSC

effects are not syntactic, and turn now to an account of comma chains in order to return to their interaction with 'wh extraction' and - more centrally - with function composition.

3.3. Comma chains, Lakoff chains, and *wh* extraction

Before returning to the specific instance of comma chains that we are calling Lakoff chains, we need an account of comma chains in general. The simplest account would posit silent lexical items *and* and *or* (and sum *and*) which have all the same properties as the overt ones. Indeed, this is the account mentioned at the outset of this paper. These items would have the same meaning and word order possibility as overt *and* and *or* - just a different phonology, and so I write them as ~~*and*~~ and ~~*or*~~ (henceforth I ignore sum ~~*and*~~ as all the same remarks hold). But this alone does not account for the constraints detailed in the introduction: in a comma chain, the last conjunct must be overt and there is consistent semantics throughout which is determined by the overt one.

Given these constraints, one account of comma chains assigns them a flat structure. But this necessitates a new type of rule making use of the Kleene star. For example, one might propose a phrase structure rule (or its equivalent in modern terms) of the form $X \rightarrow X^*_{\text{and/or}} X$, paired with an appropriate semantics; a related proposal is in Gazdar et al. 1985. I return to this in Appendix 2, where I provide evidence against this solution.

I thus assume that there are silent items ~~*and*~~ and ~~*or*~~ (and sum ~~*and*~~) with a similar syntax to their overt counterparts (and the same semantics), but I adopt a feature passing solution to account for their distribution. To this end we adopt two features [*&*] and [*v*] (and presumably a third which we notate as [*⊕*] for sum *and*) and refine the categories for overt *and* such that the category for *and* is $(X[\&]/X)/X$. This allows the overt items to

'launch' the chain (it will be the lowest conjunct) and the relevant information can be passed up. *Or* is the same but with the [v] feature instead. As to the role of these features on the case of null conjunction, we simply posit that the category of ~~*and*~~ is not exactly that of *and*. The full lexical entry for (Boolean) ~~*and*~~ (once directional slashes have been added) is: $\langle [\emptyset], (X[\&]_L X)_R / X[\&], \lambda A [\lambda B [A \sqcap B]] \rangle$. This ensures that ~~*and*~~ combines only with something as its right argument that has an overt *and* 'at the bottom', and it also passes up that feature just as does ordinary *and*. All of the same holds for ~~*or*~~ (with the appropriate semantic modification).²¹ Note that these conventions assure a right branching structure throughout for comma chains. (Here and henceforth, reference to a right branching derivation means a derivation with only application and no function composition.²²)

Moreover, recall the earlier discussion about features: if *X* itself is complex, the relevant feature will be on the result; we show empirical advantages in Secs. 5.3 and 6. So, for example, take the case of two conjoined VPs like *danced and sang*. The whole VP needs to have the [$\&$] feature because it could conjoin on its left with ~~*and*~~ (giving, ultimately, *chewed gum, danced, and sang*). But the [$\&$] feature doesn't span "VP", rather it is on the result; the category of *danced and sang* is actually $S[\&]/NP$. And this in turn means that the [$\&$] feature is propagated up; an ordinary sentence like *Lee danced and sang* is $S[\&]$. While it may initially seem strange to have the feature higher than it appears to be needed, this is unproblematic provided it is understood that any item of the form A/B can take $B[\&]$ as argument. Thus if *that* is $CP/_R S$ it can take an $S[\&]$ - which might be an S with, for example, VP conjunction within it. And it will be shown in Sec. 5.3 that having the feature appear to be 'too high' makes a somewhat unexpected (and - happily - correct)

empirical prediction. (Note that an [&] feature launched from conjoined VPs will not be on the CP; the feature stops propagating once S is reached. But see fn. 25.)

The Lakoff chain examples above - all involving *wh* extraction cases - are unsurprising given the conventions above for the passing of the | feature. The category for *and* is (X/X)/X (we ignore the [&] feature where not relevant for the discussion). Suppose X is instantiated as, for example, VP. Then the above conventions allow a version of *and* which is ((VP/LVP)|NP)/_R(VP|NP). The lexical entry for *drink* is VP/NP but it can shift (with no meaning change) to VP|NP. Thus *and* can take *drink* (here, to its right) to give *and drink* of category (VP/LVP)|NP. *bicycle home* is an ordinary VP, but can lift to VP/_R(VP/LVP). That can shift by "| passing" to give (VP|NP)/_R((VP/LVP)|NP). Then *bicycle home and drink* is a VP|NP. (There is a simpler derivation using the recursive definitions in Jacobson 2014; the interested reader can consult that.) Incidentally, these chains provide evidence not only against a syntactic CSC,²³ but also against a completely general Left Branch Condition. Obviously some principle is needed for such effects, but if it is too general it will incorrectly rule out some of the examples above. Even simple cases like (25a,b) involve a gap in a left conjunct only.

Especially interesting for our purposes is to note that the Lakoff chains in these *wh*-extraction cases allow for what I call 'mix and match'. In any two coordinated constituents, the gap can be on the left and not on the right as in (25a), (*pack up* __ *and then bicycle home*), on the right and not on the left (*bicycle home and then proceed to drink* __ as in (25c)), or on both (as in ordinary ATB cases). This is exactly to be expected given the | passing conventions adopted above. Moreover, the lowest one can contain a gap or not,

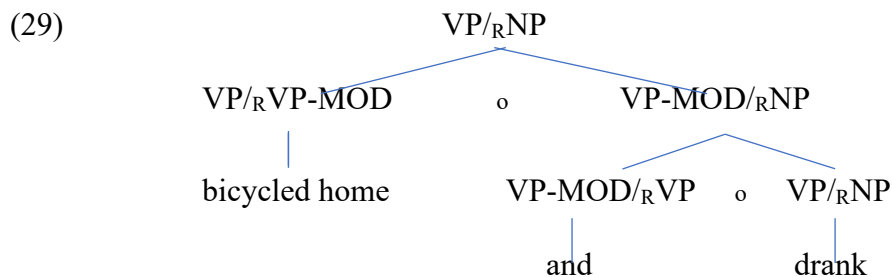
as can the highest one. When we turn to right composition (the RNR examples) and then to left composition (the "MM" examples) we see that these do not show the same freedom. For more thorough discussion of Lakoff chains in *wh* extraction, see Jacobson (to appear).

4. Rightward extraction and Lakoff chains: the lesson for function composition

Now consider Lakoff chains in the case of rightward ‘extraction’ – i.e., cases generally thought of as RNR. Here we also find that ‘mix and match’ is generally allowed. But unlike *wh*-extraction an interesting constraint emerges: the rightmost member of the chain has to contain a gap (meaning, in this context, that it has to be missing an argument). To exposit why, we begin with a good case like (28):

(28) Lee went to the store, bought __, bicycled home, and drank __ 2 sixpacks of beer.

The bottom of the chain in (28) is *bicycled home and drank* __, where we have a full VP on the right conjoining with one missing something or, put differently, with the ordinary transitive verb *drank* which is of category VP/RNP. The expression *bicycled home and drank* can combine by function composition. For expository convenience, we use the abbreviation VP-MOD for VP/LVP (this is the category of, e.g., *and chews gum*). Note moreover that an ordinary VP like *bicycled home* can lift to VP_R(VP_LVP) (e.g., VP_RVP-MOD). In the illustration below of the derivation we begin with the lifted category of *bicycled home* to save space, and so the lowest VP here is analyzed as follows (we suppress the & feature):



Given the derivation above of *bicycled home and drank*, two things are possible as we put together bigger expressions. This expression can combine with a VP on its left in exactly the same way - using either overt *and* or, in the cases of interest here, ~~*and*~~. This for example is what happens in (30), where *bicycled home and drank* __ combines with *found his bicycle* in the same way as happens at the bottom of the chain as shown in (30).

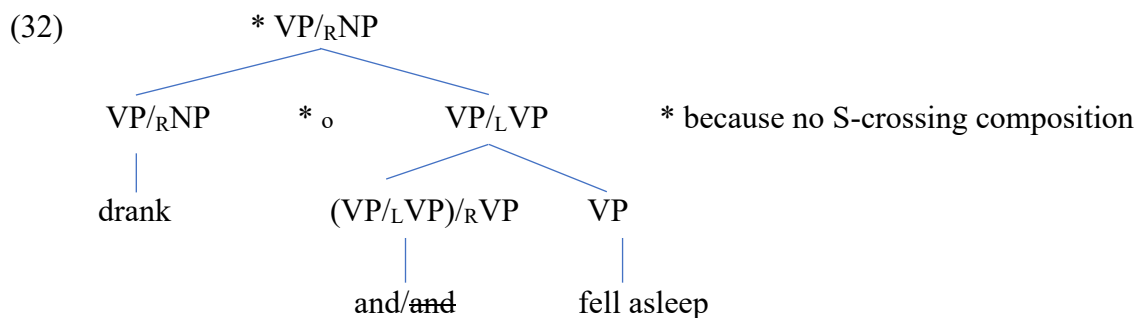
(30) Lee bought __, found his bicycle, bicycled home, and drank __ 2 sixpacks of beer. Alternatively, *bicycled home and drank* __ can combine with another VP/RNP by ordinary coordination of likes; this is exemplified in (28) where *bicycled home and drank* __ conjoins (by ~~*and*~~) with *bought* __. (This also happens in (30) at the top of the chain.). So, once the chain is launched, we can keep mixing expressions with a gap with ones without.

We have dealt here only with combinations of VP/RNP (transitive verbs, whether simple or complex) and VP, but we should also find this with S/RNP and S. And we do:

(31) George loved __, Paul hated __, Ringo was just like 'whatever', and John insisted on performing __ that silly song about a yellow submarine.

While this looks so far just like the situation with *wh* extraction, the account here makes a striking prediction. We see that a VP on the left can combine with a VP/RNP on the right (with *and* or ~~*and*~~ as intermediary) and the result is VP/RNP. But there is no way for an 'ordinary' VP on the right to combine with a VP/RNP to the left so as to give VP/RNP

- the category needed in RNR (and "Heavy NP Shift"). For this would not be function composition - it would have to be S-crossing composition. For example, take *drank* which is of category VP/RNP and *fell asleep* of category VP. *and/and* can of course take the VP on its right to give VP/LVP. But there is no way for *and/and* *fell asleep* to take a VP/RNP to its left to give VP/RNP) ; note that the use of * indicates an impossible composition:



Notice that S-Crossing composition gives exactly the above. The VP/LVP *and/and* *fell asleep* would take to its left the VP/RNP *drank* yielding a VP/RNP. But we stress that it is not true function composition and so is precluded under the view of categories here.¹⁷

Is this prediction correct? Indeed it is; compare (33a,b) to (33c) and (31) to (34):

- (33) a. *Lee drank __ and fell asleep 2 sixpacks of beer.
 b. *Lee went to the store, bought __, drank __ and fell asleep 2 sixpacks of beer.
 c. Lee went to the store, bought __, bicycled home and drank __ sixpacks of beer.

- (34) *Paul loved __, George hated __, John insisted on performing __ and Ringo was just like 'whatever' that silly song about a yellow submarine.

In sum, while RNR cases do allow 'mix and match' in a Lakoff chain once it is launched, it has to be launched by having the 'smaller' item (the one of category A/RB)

being the rightmost one. This contrasts with the case of leftward *wh* type extraction. By way of a complete minimal pair, compare (25b) to (35):

(35) *Lee will buy__, drink__, and still not fall asleep 2 sixpacks of beer.

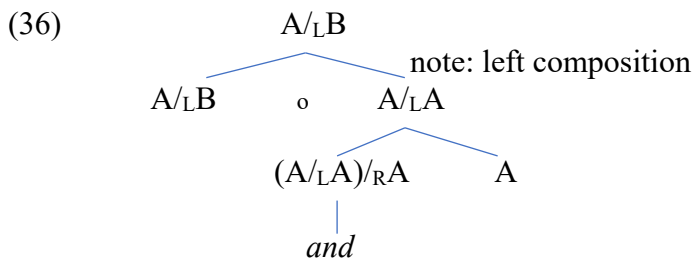
Hence this phenomenon also provides new evidence that the two require different conventions (beyond those already noted in past literature, see e.g., McCloskey 1986).

5. Leftward function composition chains

5.1. The predicted and actual existence of left looking Lakoff chains

We arrive to the centerpiece of this paper: the existence of Lakoff chains where the function composition schema is instantiated by 'left composition'. As noted above, a chain with a rightmost 'filler' is launched by something of the form A and A/RB , thus we find (among many others) chains of the general form: $[A \text{ and } [A/RB \text{ and } A/RB]] B$. The mirror image - in terms of the string order - would be $B [A/LB \text{ and } [A/LB \text{ and } A]]$ (The bracketing is not mirror image due to the asymmetry of *and* and *and*). This raises two questions. Would we expect to find such cases? And do we? The answer to both is yes.

Turning first to the question of whether we should expect these, we can show that a left-looking Lakoff chain can be launched as shown below:



This can then conjoin with another A/LB by the ordinary coordination of likes, where that conjunction is either via an overt *and* or an instance of *and*; the latter gives rise to a Lakoff

chain. Once that is done the string order will be exactly what is noted above: $B [A/_LB [and A/_LB and A]]$ - or, to instantiate this with actual categories: $NP [S/_LNP [and S/_LNP and S]]$. Another instantiation would be $TV [VP/_LTV [and [VP/_LTV and VP]]$ (recall that $VP/_LTV$ is the category of an NP lifted over transitive verbs; its full category is $(S/_LNP)/_L((S/_LNP)/_RNP)$ where this is a valid lift of an NP). And recall that (leaving aside sum *and*) the NPs would have to lift as Boolean $[[and]]$ is not defined for individuals.

The next question is: do we get these? The answer is yes - these are exactly the Maxwell/Manning cases noted in the introduction. Simplified versions are (37) and (38);

(37) Barry $[S/_LNP$ plays the dulcimer ~~and~~ $[S/_LNP$ does magic tricks [and $[S$ Patty plays the flute]]]].

(38) Barry $[TV$ plays $[VP/_LTV$ the dulcimer ~~and~~ $[VP/_LTV$ the standup bass [and VP does magic tricks.]]]].

Because the syntax and the semantics involve parallel computations, the semantics of these and all subsequent examples work out as expected. By way of illustration take (37). The lowest *and* (the overt one) is propositional *and* (here we need to give the semantics intensionally as Π is defined for two arguments of type $\langle s, t \rangle$). Hence *and Patty plays the flute* is $\lambda p_{\langle s, t \rangle} [p \Pi \text{patty-play-flute}']$. This function composes with *does magic tricks* - of type $\langle e, t \rangle$, to give $\lambda x_e [\text{does-magic-tricks}'(x) \Pi \text{patty-play-flute}']$. This conjoins with *plays the hammered dulcimer* (in two steps since ~~and~~ is Curry'ed) by the ordinary conjunction of likes. The resulting semantics for *plays the dulcimer, does magic tricks, and Patty plays the flute* is $\lambda x_e [\text{plays-the-dulcimer}'(x) \Pi \text{does-magic-tricks}'(x) \Pi \text{patty-}$

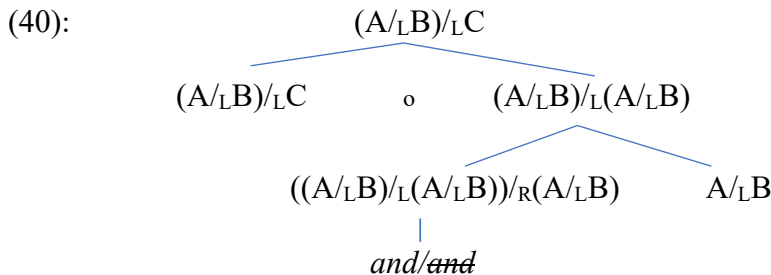
play-flute']. This function is ultimately applied to [[Barry]]. (One might think that the M&M examples instead involve ellipsis; see Appendix 3 for arguments against this.)

The semantics of (38) is also straightforward. *The standup bass* lifts from its initial e-type meaning to a meaning of type $\langle\langle\text{eet}\rangle,\langle\text{et}\rangle\rangle$; in particular its meaning is $\lambda R_{\langle\text{e},\langle\text{e},\text{t}\rangle\rangle}[\text{R}(\text{the-standup-bass})]$. The meaning of [[and does magic tricks]] is $\lambda P_{\langle\text{e},\text{t}\rangle}[\text{P} \sqcap \text{does magic tricks}]$, which is of type $\langle\text{et},\text{et}\rangle$. But this combines not with an $\langle\text{et}\rangle$ meaning but with something of type $\langle\text{eet},\text{et}\rangle$ via function composition, yielding $\lambda R_{\langle\text{eet}\rangle}[\text{R}(\text{the hammered dulcimer}) \sqcap \text{does magic tricks}]$. The same thing happens further up. Hence the meaning of *the dulcimer, the standup bass, and does magic tricks* is (39):

$$(39) \quad \lambda R_{\langle\text{eet}\rangle} [\text{R}(\text{the standup bass}) \sqcap \text{R}(\text{the dulcimer}) \sqcap \text{does magic tricks}]$$

[[plays]] eventually fills in the R slot, and we get the intersection of playing standup bass, playing dulcimer and doing magic tricks, exactly as we want. We will henceforth omit the semantic composition for each case as the semantics is predictable from the syntax.

We also predict that there are more complex Lakoff chains. Note that an expression of any category $A/_LB$ itself could combine with something smaller to its left such as $(A/_LB)/_LC$ in exactly the same way that was illustrated in (36):



To put this all together, take the tree in (36) and imagine that it is the expansion of the rightmost node in (39) (the node A/LB). This yields chains like (41), shown first in the abstract and then instantiated with actual categories, and finally actual words:

- (41) $[(A/LB)/LC \quad \text{and} \quad [A/LB \quad \text{and} \quad A]]$
 $[(S/LNP)/LTV \quad \text{and} \quad [S/LNP \quad \text{and} \quad S]]$
the dulcimer ~~and~~ does magic tricks and Patty plays the flute

We cannot show that this exists simply by placing the TV *plays* to the left, as that has a perfectly good different parse - the very one in (37) where *plays the dulcimer* can be analyzed as a VP conjoining with *does magic tricks*. Nonetheless we can show that cases like (40) exist by conjoining *the dulcimer* to the left with another raised object by the ordinary conjunction of likes. And indeed this is good:

- (42) Barry plays the bass, the dulcimer, does magic tricks and Patty plays the flute.

The question was raised earlier as to whether the 'growth' needs to be incremental: do we, e.g., have chains of the form *(lifted) NP, (lifted) NP, and S*? The answer is yes:

- (43) Barry plays the hammered dulcimer, the standup bass, and Patty plays the flute.
NP TV $[(S/LNP)/LTV \quad [\text{and} \quad [(S/LNP)/LTV \quad \text{and} \quad S]]]$.

There are in fact two ways to get this. One simple analysis involves using the "Geach" rule (**g**) rather than direct function composition (see fn. 15). Recall that *and S* is a wellformed S/LS . We need then only two applications of **g**. The first maps this to $(S/LNP)/L(S/LNP)$ (a VP modifier). This then maps to a lifted object modifier $((S/LNP)/LTV)/L((S/LNP)/LTV)$. Hence *and Patty plays the flute* takes *the standup bass* (a lifted object) as argument. From there we have coordination of likes (*the hammered dulcimer* is also a lifted object). But

there is also an analysis using just function composition. *the hammered dulcimer* lifts to category $S/L(S/RNP)$. Since *and Patty plays the flute* is S/LS , the two can compose to give *the hammered dulcimer and Patty plays the flute*, of category $S/L(S/RNP)$. *the standup bass* lifts in the same way and we have coordination of likes in the expression *the standup bass, the hammered dulcimer and Patty plays the flute*. Recall that *Barry plays* can be analyzed (with lifted *Barry*) as an expression of category S/RNP and that thus is the argument of the complex expression above.

The actual (single) example in Maxwell and Manning 1996 involved not just a simple object but a "fancy" one of the type analyzed in Dowty 1988:

- (44) John wanted to study medicine when 11, law when 13, and to study nothing at all when 18.

(Incidentally, their analysis of these has virtually nothing in common with the analysis here nor did they discuss the phenomenon in any generality.) These examples are fun but really add nothing new. *medicine when 11* can compose up in the way shown earlier for (16) to be something wanting a TV to its left to give a VP, so the rest is the same as the case of an ordinary object which lifts over TVs. And unsurprisingly - parallel to the "Barry" sentence in (42) except with 'fancy' objects - we find more complex chains like (45):

- (45) Laura teaches syntax in the fall, compositional semantics in the spring, goes to music workshops in the summer, and she spends each sabbatical writing a book.

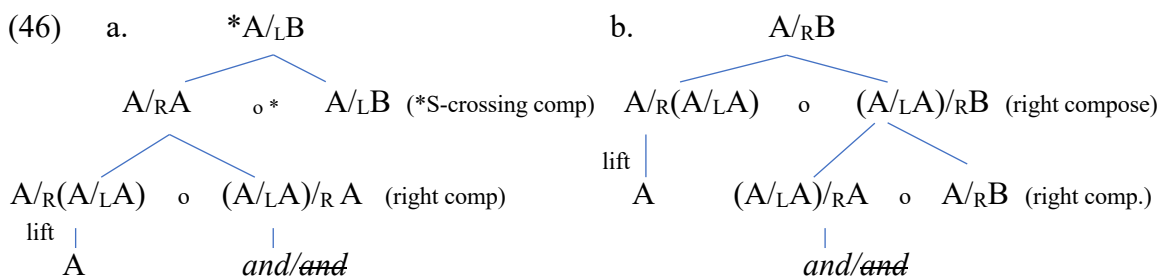
5.2. The "Grow Only Rightward" phenomenon: No S-Crossing Composition

Recall that in the case of a Lakoff chain looking for something to its right, the rightmost one had to be the smallest, but otherwise any kind of mix and match is possible.

The left looking case is different. We have seen how larger expressions on the right (lower down) can combine with smaller ones to their left - such cases can be 'launched' by having the bottom of a chain being an expression of the form $A/_LB$ and A where the full expression itself has category $A/_LB$. (As discussed above - a chain can also be launched by a string of the form $(A/_LB)/_LC$ and A yielding an expression of category $(A/_LB)/_LC$). But there is an interesting prediction: an expression of, e.g., category $A/_LB$ should not be able to combine with something larger to the left ("larger" means a category like A as opposed to A/B).

To exposit, let us try to compose an expression of the form A and ~~and~~ $A/_LB$ - either at the beginning (rightmost, lowest) part of the chain (which would involve overt *and*) or anywhere higher up (using ~~and~~) to give an expression looking for something on its left (i.e., an $A/_LB$). (If this can't be done, it follows that we also can't get A and ~~and~~ $(A/_LB)/_LC$.) Consider what it would take for *and* (or ~~and~~) to combine on the right with an expression of category $A/_LB$. First, we could use the version of *and*/~~and~~ of category $((A/_LB)/_L(A/_LB)/_R(A/_LB))$. This is fine but just involves coordination of likes, so the rightmost category is the same as the left conjunct. Let us try instead to use the version of *and*/~~and~~ of the simpler category $(A/_LA)/_RA$. Can this combine with an $A/_LB$ to its right to give $(A/_LA)/_LB$? This, for example, would allow strings of the form S and VP (i.e., $S/_LNP$) of category $S/_LNP$ at the bottom - or as part of - a left looking Lakoff chain. It would also allow a string VP and $VP/_LNP$ (the latter being the category of lifted objects) where that string has the category $VP/_LNP$? If there were S-Crossing composition the answer is yes. But under the view of categories here the answer is no.

We might still try one more tack a different tack, which is as close as possible to the mirror image of what happens in the right looking case. This would be to have *and* combine with its arguments in a left branching fashion. After all that is allowed by function composition in just the same way that we have a left branching derivation of, e.g., *Lee tells lies* (as discussed in Sec. 2.8.3). The problem here is that the next step once again would require S-crossing composition. We can illustrate this, comparing the left looking derivation in (46a) with the right looking one in (46). The categories shown in bold are the ones we are ultimately trying to conjoin to get result being aimed for, but the category at the top requires S-crossing composition, so (46a) does not exist:



There are other possible things one might try given combinations of Lift and other operations, but since Lift is order preserving it won't matter; there is no way to get a larger expression to the left of a smaller one, not only at the beginning of the chain but also anywhere higher up. It is thus predicted that the only way to get leftward Lakoff chains is as shown in Sec.5.1. Things can grow rightward, but they cannot shrink.

Strikingly, this prediction is correct. Compare (47a) to (47b):

- (47) a. Sarah teaches Lexical Semantics in the fall, Compositional Semantics in the spring, and plays the violin in her spare time.

- b. *Sarah teaches Lexical Semantics in the fall, plays the violin in her spare time, and Compositional Semantics in the spring.

(b) is good only on the pragmatically bizarre reading in which Lee plays Compositional Semantics. Similarly (48b) has only the strange reading where Barry plays treehouses.

- (48) a. Barry builds dulcimers, treehouses, and plays the standup bass.
b. *Barry builds dulcimers, plays the standup bass, and treehouses.

5.3. The surprising reading: features on result categories

The same point should hold in the case of chains of Ss followed by VPs. At first glance, it appears that this generalization does not hold in that case; (49a) is the expected order, but (49b) - where we flip the VP and S - is surprisingly good:

- (49) a. Sally teaches semantics, studies math, and she plays the crumhorn.
b. Sally teaches semantics, she plays the crumhorn, and studies math.

But on closer look, this turns out not to be a counterexample. In fact, it involves a different and surprising analysis - the very same analysis which allows for the pragmatically bizarre readings of (47b) and (48b). This is that *plays the crumhorn* and *studies math* are conjoined VPs which take *she* as subject to give an S. Thus the structure is (50b) rather than (50a); (50a) would require S-crossing composition:

- (50) a. Sally teaches semantics ~~and~~ [[_S she plays crumhorn] and [_{VP} studies math]].
b. Sally teaches semantics ~~and~~ [_S she [_{VP} plays the crumhorn and studies math]].

It is easy to show that (50b) is the right structure simply by using a different subject:

- (51) Sally teaches semantics, Lee plays the crumhorn, and studies math.

We cannot understand this as saying that Sally studies math; it is Lee who studies math. That is what we would expect from the structure in (50b), whereas under (50a) *Sally* is the subject of the lower VP, and the separate sentence *Lee plays the crumhorn* would have nothing to do with the interpretation of the math-VP. Thus the Grow Only Rightward prediction is born out, which in turn supports the hypothesis of no S-crossing composition, as we would expect under the view of syntactic categories here.

But (50b) and (51) present a new mystery: shouldn't they also be bad? Isn't the overt *and* too low to support ~~and~~ higher up? (For reasons which will become clear in the next section, let us call this 'too high ~~conj~~'). Actually not. Recall the hypothesis that features are always encoded on the ultimate result category. That the [&] feature is on the result category makes exactly the right prediction here. The full (50b) is:

- (52) Sally teaches semantics ~~and~~ [S[&] she [S[&]/NP [S/NP plays the crumhorn] and [S/NP studies math]]].

We should also find this when the overt *and* conjoins lifted objects. While perhaps awkward, (53) also seems good with the right prosody. These are best with have rising intonation and a pause before what in writing is commas (including the last Oxford comma). And we correctly predict that these can also have an *or* reading as in (54):¹⁹

- (53) Barry will play the crumhorn, Patty will play the flute, and the Irish whistle.
- (54) In order to have enough courses, either Lee should teach Pragmatics in the fall, Sandy teach Intro in the spring or (else) develop a new course for January term.

And we should also find this with conjoined lifted objects and VPs - and we do. Recall that (47b) and (48b) had pragmatically strange readings: these are exactly the readings here. The following remove the pragmatic anomalies:

- (55) a. Sarah conducts an orchestra, plays the violin and occasionally the viola.
 b. Sarah deserves tenure. She wrote three papers while on sabbatical, taught syntax the following fall, and semantics that spring.

There is one further interesting prediction. Because the [&] and [v] features which seem 'too high' must be on a result category, we also predict that they will stop at an S node. Thus *and* will not license silent comma chains higher up if it is an embedded S and the silent conjuncts are higher. This appears correct; compare (55a) and to (56):

- (56) *Sarah conducts an orchestra, claims that she plays the violin and occasionally the viola.

6. Additional confirmation: the [v] feature and the distribution of *either*

There is a substantial literature on the surprising distribution of *either* - where it appears at first glance to sometimes be 'too low' (with respect to the item *or* which licenses it) and other times 'too high'. The former is discussed in, e.g. Hendriks, 2004 and den Dikken, 2006, and is illustrated by the following:

- (57) Barry will play either the dulcimer or (will) do magic tricks.

The latter was first noted by Larson 1985 in a seminal paper drawing attention to *either*, and is illustrated by (58), see also Schwarz 1999:

- (58) a. Either Sarah will teach syntax or will develop a course on semantics.
 b. Sarah will either teach syntax or semantics.

The literature especially on too high *either* is complex and the judgments not always clear and not always agreed on. It would thus be premature to claim here to have a full account of the too high case. That said, the basic distribution of both too low *either* and too high *either* fall out effortlessly from the [v] feature distribution motivated here for comma chains combined with the hypothesis of the feature being on the result (although further refinements might be needed for the too high case). Put differently, there is a striking commonality between too high *either* and too high ~~*conj.*~~. And too low *either* follows in exactly the same way as does the existence of left looking Lakoff chains.

To see the initial intuition regarding the too low case, note that in (57), for example, *either* is supported by the "strange" looking disjunct *the dulcimer or (will) do magic tricks*. But that disjunct should no longer strike us as strange given function composition. It is exactly the disjunct that launches a left looking Lakoff chain like *the standup bass, the dulcimer, or (will) do magic tricks*. Elucidating further, assume that *either...or* is not a single discontinuous constituent; rather, *either* needs to find *or* somewhere. This is easy to accomplish with the feature passing mechanisms used here: assume that *either* takes an expression with the [v] feature and gives back the same expression without that feature, hence its category is $X/X[v]$. That it does indeed swallow up this feature can be shown as follows. Note first that while some people do not like *either* with more than one disjunct, many do, and so we predictably can get this in comma chains as in (59):

(59) He'll either dance, sing, or jump.

But a lower *either* will remove the possibility of a higher one as in (60):

(60) *He'll either dance, either sing or jump.

And because comma chains require the $[v]$ feature on the topmost member of the chain, (61) is also predictably bad because *either* has removed the $[v]$ feature: (the corresponding case with overt *or* is predictably fine, as overt *or* does not need to find the $[v]$ feature):

(61) *He'll dance, either sing or jump.

We assume that the semantics of *either* is the identity function although it does strengthen the exclusive implicature of *or* for reasons left open here.

Hence given the feature passing conventions adopted earlier for $[v]$, nothing more needs to be said to license too low *either*. As noted above, its existence follows directly from function composition. Consider (57). Here *the dulcimer or will do magic tricks* is put together just as the last two conjuncts/disjuncts are in any left looking Lakoff chain, where the relevant feature is inherited on the result by the way function composition works. *or does magic tricks* is of category $VP[v]/_L VP$ (we show the $[v]$ feature on the full VP result here but recall it will actually be on the S-part of the category $S[v]/NP$). *the dulcimer* can lift over transitive verbs to be of category $VP/_L TV$. When these compose the result is $VP[v]/TV$. Recall that the category of *either* is $X/X[v]$, but recall further that when we write $X[v]$ that is shorthand for any category where $[v]$ is on the ultimate result. In more detail, *either* takes as argument any expression whose category has $[v]$ on the result, as is the case for *the dulcimer or will do magic tricks*, whose full category is $(S[v]/_L NP)/_L ((S/_L NP)/_R NP)$. So in (62) it is no surprise that *the dulcimer or will do magic tricks* can be argument of *either* (which would remove the feature). Or of this could just instead combine (again by function composition) with (lifted) *the standup bass* to give a Lakoff chain, and *either* can be introduced there (or never):

(62) Barry will play either the standup bass, the dulcimer or will do magic tricks.

The case of too high *either* - as in (58) - provides striking confirmation for another crucial claim here: that CG allows features to be encoded on result categories (as we would expect for theoretical reasons). As noted above, the case of too high *either* is the same as the case of too high ~~*either*~~ - which is what accounted for the surprising goodness of the examples in Sec. 5.3. Thus in (58b) *either* is at the VP level but *or* combines two (lifted) NPs. In (58a) *or* is the disjunction of two VPs, yet *either* shows up above the whole S. This too follows from the analysis of feature passing here. Recall that *Barry play the dulcimer or the bass* is of category S[v]. *Either* can then combine with this and 'swallow up' the [v] feature. Similarly it can combine in (58a) with the S[v]/NP (returning S/NP).²⁵ The conclusion here is that 'too low' *either* is licensed in a fashion exactly analogous to what gives left looking Lakoff chains, and 'too high' *either* is licensed in a fashion exactly parallel to the 'surprising' examples in Sec. 5.3.

We leave this discussion by noting another intriguing parallel between too high *either* and too high ~~*either*~~, although this is one for which we do not have a definitive account. A key observation of Larson 1985 was that (63a) has a narrow scope de dicto reading, but (63b) and (63c) do not:

- (63) a. They are looking for either a semanticist or a phonologist.
 b. They are either looking for a semanticist or a phonologist.
 c. Either they are looking for a semanticist or a phonologist.

While having no full account, we might speculate that for some reason the [v] feature stops going up when an intensional object like *a semanticist or a phonologist* is object of an intensional verb. It would follow that the sister of *either* lacks the needed feature.

If this is correct, we expect to find intensional objects conjoined with *or* (with a de dicto reading) to also not support a higher θ . Happily, this seems correct. Imagine a fledgling Linguistics Department with many needs, but also knows they will be given minimal resources. Suppose they hope that at least they can move into a new building, or that they can have a dedicated search for a semanticist or a dedicated search for a phonologist. Now take the same scenario but where their hope is to get a new building, or be allowed to conduct a disjunctive search (just looking for the best candidate who is a semanticist or phonologist). We can report the first (with the right prosody) but not the second scenario as follows:

(64) The hope is to get a new building, search for a semanticist or a phonologist.

It is striking that the distribution of 'too high' *either is* exactly the same as the distribution of 'too high' silent conjunctions in comma chains; neither supports the de dicto reading of the lowest disjunct. (It might also be that the remarks in fn. 22 extend to account for the missing reading in (64); the hope then would be that *either* is also sensitive to the finer grained feature information. We leave this for interested readers to pursue.)

7. Conclusions

I have made three main suggestions embedded in a CG framework. The main one is that 'function' categories correspond to functions taking two strings as argument to give a third. Once we add in Lift and Function Composition, this view predicts that only certain

types of lifts and compositions are available - just those which literally correspond to those operations. Of greatest relevance here is that this precludes the possibility of what we are calling S-crossing composition. Second, I have proposed an analysis of comma chains which gives them a right branching structure and accounts for their basic distribution and interpretation. The third is that features on function categories are on the result. I have also argued against a syntactic CSC (hardly a new position here). Taken together, these claims account for a number of new and not new facts: the existence of RNR Lakoff chains and - more strikingly - the fact that these must have a gap in the rightmost conjunct; the existence of left looking Lakoff chains (i.e., the MM cases) and the Grow Only Rightward phenomenon, the (at first glance) surprising possibility of the 'too high' *conj* cases, and (with some open questions) the existence of both too high and too low *either*.

Appendix 1. More on Infixation

A1.1. The category of ordinary transitive verbs.

It was speculated in Sec. 2.7 that a 3-place verb like *give* (as in *give the bone to the dog*) has as its category $((S/LNP)/_iNP)/_RPP$. This raises the question of how to add or revise the word order rules given earlier in (10); the answer has relevance to the analysis in Sec. 4. Assuming that the post-verbal NP is an infix in 3-place verbs in general, we need to at least add to our stock of word order rules a rule: $...((S/X)/NP)/Y... \rightarrow ...((S/X)/_iNP)/Y...$ Assume for the moment that Y here stands for any category but cannot be empty.

But it is reasonable ask whether we really need to specify that this holds only for 3-place verbs. Put differently, this raises the question of what is the actual category of a

simple transitive verb such as *likes*. The most general statement of the rule adding an I-feature is for it to map any item of category $\dots((S/X)/NP\dots$ to $((S/X)/_I NP)\dots$. That would mean that even a simple transitive verb takes its object as an 'infix', but in the case where there is no other argument, the infixation is vacuous. Since the infixation point is to the right of the verb, 'infixation' yields the same result as does place the argument to the right. (This is parallel to morphological infixes which sometimes look like prefixes.) But recall that the analysis of RNR in Sec. 4 assumed that ordinary transitive verbs are of category $(S/_I NP)/RNP$. But since the rules are simplest if 'direct objects' are always infixes (even where vacuously so) one might wonder whether treating transitive verbs as having an R slash rather than an I slash was justified.

I believe that this is justified and/or at least a justifiable expository simplification for three reasons. First, it certainly involves no major complication to restrict the infixation slash to only 3 place verbs. Second, everything of category $\dots ((S/X)/_I NP \dots$ also has a version where that I-slash is an R-slash; this is what gives the Heavy NP Shift cases. As such, there is no harm in working with the R-slash version of transitive verbs. ("Heavy NP Shift" would be vacuous in the case of simple transitive verbs). In fact, using the R-slash version plays a role here only in the analysis of Right Node Raising (note that it plays no role here in *wh*-extraction cases), and RNR that has the same basic prosody as does Heavy NP shift. So using the same mechanism for both makes some sense. Finally - and most importantly - even if we were to make the assumption that ordinary transitive verbs always take objects by vacuous infixation, I suspect that everything in Sec. 4 using the simplification there would carry over directly to a full theory with infixation spelled out.

A1.2. Infixation and crossing composition

We have been careful not to claim that the theory of function composition here precludes any kind of "Crossing" or "mixed" composition; rather, we focused on the fact that it precludes S-Crossing composition. Depending on how exactly all of the details are formulated, it is conceivable that some kinds of crossing composition would be allowed. Consider the case of *believe* __ *lost* which we have shown cannot be VP/LNP (although it can be VP|NP). But it could conceivably be put together to have yet some other category. After all, there is a function F' -*believe*; which is a function from strings of category S to strings of category VP. And there is a function derived by composing F' -*believe* with F' -*lost*. Whether that composition is actually allowed in the grammar depends on how infixation point inheritance is encoded in categories (see Jacobson, Li, and Robertson in preparation). In any case, though, that composition would not yield VP/LNP; it would yield VP/I_{NP} (where *believes lost* would take *Joe* as infix to give *believes Joe lost*). If that is allowed, it is harmless.

Appendix Two: Comma chains as flat conjunctions?

There have in the literature been proposals to the effect that chains of silent conjuncts/disjuncts involve a 'flat' structure (e.g, Gazdar et al 1985). For example, one might propose something like the following phrase structure rule schema for these:

(65) $X \rightarrow X^* \text{ and } X$ (similarly for *or*)

Note that * here is the Kleene star - the interpretation of this means any number of expressions of category X. (Hence this abbreviates an infinite number of 'ordinary' rules.) This can have an associated semantics such that the relevant operation is that which is

dictated by overt conjunction. Thus in the case of *and*, the meaning of the mother is the generalized meet operation of the daughters.

This admittedly has a striking advantage: there is no need for any kind of feature passing to keep track of the semantics. Whether the last part of the chain is *and X* or *or X* (or sum *and X*), the chain will be interpreted consistently throughout. (It also would render moot the discussion in fn. 22.) That said, there are both theoretical and empirical reasons for rejecting this or any similar schema. As to the theoretical reasons, note first that rules of this general form are not compatible with any theory countenancing only binary rules as in Minimalism or the version of CG here. Of course the hypothesis that all rules are binary could be wrong (note, though, that it is not a stipulation but a natural assumption in CG). But aside from that, this requires the incorporation of the Kleene star. Depending on other theoretical assumptions, this would be a new device.²⁶ There are also empirical reasons to reject this solution. The most obvious and relevant one here is that it provides no ready account of the left looking Lakoff chains (the M&M cases). (A referee suggests that those actually involve ellipsis; we turn to that in Appendix 3.) It should be stressed again the left looking Lakoff chains require nothing extra in a theory with function composition; this phenomenon falls out automatically. The very device which also gives us the RNR cases gives the left looking Lakoff chains, and we would be surprised if these did not exist. Under a flat analysis something extra is needed. And even if we could find a different mechanism for the M&M cases, it needs to account for the Grow Only Rightward Phenomenon (again see Appendix 3 on ellipsis).

A second empirical problem with the flat structure is that there is no obvious way for it to allow for the "surprising reading" cases in Sec. 5.3 (examples (51) and (53)-(55)). Under the flat analysis, the *and* in these cases genuinely is too low; it could not be part of a chain licensed by any kind of rule of the general form $X \rightarrow X^* \text{ and } X$ (or any modern equivalent of this). Recall that these examples follow immediately from the right branching analysis combined with the hypothesis that the feature is on the result category. And while one might try ellipsis for the basic M&M cases (but see Appendix 3), that would not extend to the cases in 5.3, as the interested reader can verify.

Finally, consider the behavior of *but* in comma chains. In particular compare (66a) to (66b) which has overt instances of *but* throughout:

(66) a. Patty plays the flute, doesn't really like blues, but does love *Matchbox Blues*.

b. Patty plays the flute, but doesn't really like blues, but does love *Matchbox Blues*.

(66b) shows that it is perfectly sensible for each conjunct to have a 'but' reading - that reading implicates that flute players usually like blues. Despite the fact that this is pragmatically sensible, (66a) has no such reading - it suggests no particular connection between the first and second conjunct. Of course these will usually be in a context where there is some such connection, but what that connection is can be many different things and is not hardwired into the semantics. It is widely agreed that *but* has the same truth conditions as *and*, but with an extra implicature. The chain as a whole has *and* semantics, with crucially, the *but* implicature only for the last conjunct, where we get the obvious and unsurprising implicature.

Note that this behavior is automatic in the right branching analysis, which allows the composition (including implicature) to be put together conjunct by conjunct. Thus assume that there is no silent *but*. This means that the category of (overt) *but* - like that of *and* - is $(X[\&]_L X) / X$. Then this launches a comma chain with *and* semantics throughout (licensing *and* higher up). The one overt instance will nonetheless have the *but* implicature as it is interpreted like any other *but*. It is unclear to me one could possibly formulate this under the flat analysis. One needs a generalized 'meet' or 'join' (or 'sum') operator for the semantics of the whole chain under a phrase structure rule schema like (62), where meet, join, or sum is determined by the overt member. But there is no obvious way to fold in the extra implicature that the last *but* contributes. This is because it has no compositional contribution of its own; it merely determines the semantics of the whole chain.

Appendix Three: Ellipsis for the MM examples?

A referee suggests that the left looking Lakoff chains perhaps are not a matter of coordination using leftward function composition, but are instead an instance of ellipsis. This was paired with the idea that comma chains involve a flat structure which would necessitate some such analysis, since function composition would be unavailable. But even if one rejects the flat analysis of comma chains, there remains the possibility that the M&M examples involve ellipsis (rather than function composition), so it is worthwhile evaluating this possibility. As it turns out, there are reasons to be suspicious that this is ellipsis.

First, we need to say something about why it that ellipsis is allowed in this configuration. (Note again that as long as one grants that the right branching structure is motivated - and that there is function composition - then nothing is needed for the function

composition analysis of left-looking Lakoff chains.) More seriously, ellipsis by itself does not account for the Grow Only Rightward phenomenon. To that end, the referee suggests that ellipsis is subject to a constraint that the antecedent must be in the immediately preceding clause. This is quite clever and delivers the desired result. But - while there is often a preference for closer antecedents - that is certainly not a general property of ellipsis. Even the rather fussy case of Sluicing is not subject to such a constraint as in (67), and we can even construct an 'at a distance' case in a Lakoff chain as in (68):

(67) I've decided to buy one of those hats. I'll see how much money I have and then I'll figure out which.

(68) At the graduation ceremony, let's announce that someone in the class has just won a huge prize, have a big drumroll, and then we can announce who.

While (67) may be awkward, the case of (47b) and (48b) are different. These simply cannot pick up their antecedent from two conjuncts earlier and thus force a pragmatically deviant reading. Take, e.g., (48b): *Barry builds dulcimers, plays the standup bass, and treehouses*. This has only the bizarre reading where Barry plays treehouses. If this were just a processing preference, that should be overridden by pragmatics, just as is the case for (69):

(69) Lee hid my ring somewhere. Sandy will talk to him and will then tell me where.

Grammatically there is no reason why *will then tell me where* could not be picking up the location of where Sally talks to Lee (on a Sprouting reading). If there were simply a strong principle forcing the immediately preceding clause to be the antecedent, that should be the reading for (69). But indeed here the pragmatics clearly wins out, and the 'at a distance' reading of (69) is quite smooth. Thus having to pick up the antecedent from the

immediately preceding clause is neither a general property of ellipsis nor a general processing strategy, and so would need to be stated idiosyncratically for ellipsis in comma chains. (In fact, it is also well that in ellipsis in general the antecedent can follow, as in '*While I don't know who, I heard that she nominated someone*' But if this were allowed in comma chains we would predict that *Barry played the dulcimer, the flute, and Patty played the crumhorn* could mean that Patty played the flute. In sum, comma chains just do not behave like other types of ellipsis.

Endnotes

* This paper began at a workshop on Coordination at ESSLLI in 1996, organized by Ivan Sag, where I first heard about the Maxwell/Manning facts, and realized that they had a natural analysis in CG using left composition. I buried the paper until the EmmonFest for Emmon Bach's 85th birthday in 2014, when I realized that if one took seriously the notion of syntactic categories as corresponding to functions in $\langle \text{string}, \langle \text{string}, \text{string} \rangle \rangle$ it made an interesting predictions (the "Grow Only Rightward" prediction) which - to my delight - turned out to be true. The paper has grown and been refined since then due in part to input from others and other audiences. I thank the audiences at the above two workshops, as well as audiences at MIT, Yale, and the Edinburgh PullumFest in 2023, where more recent versions have been presented. I also am deeply grateful to Alan Munn for his (correct) intuition that this might have a connection to the distribution of *either*. In addition, I especially thank Mark Steedman, Adam Przepiórkowski and two anonymous referees for *Linguistic Inquiry* for many substantive comments which have led to great improvements. And finally I thank my friends Patty and Barry Simon whose eclectic musical and other

talents have enabled me to easily construct just the right sorts of examples (which in all but two small cases are true), and for giving me permission to use their names in the examples.

¹ While these are not well known and have not been much explored, Adam Przepiórkowski (personal communication) has collected the following naturally occurring examples:

- (i) Her father is a local businessman, author, and had recently made an obnoxious foray into rightwing politics. (from Eliza Clark, *Penance*, Harper Collins, 2023)
- (ii) It was not a day for sailing – it was cold, wet and the afternoon already well advanced. (from L.C. Tyler, *Too much of water*, Constable, 2021)

² In accordance with reasonably standard terminology in CG (and many other places) I use the term NP; the reader can substitute DP if preferred. Note too that I call these "NP", but following the semantics in Partee and Rooth 1983 these cannot have meanings of type e but have to be lifted to generalized quantifiers (type $\langle\langle e, t \rangle, t \rangle$). Given other assumptions in this paper (see Sec. 2.8) this means their category is not NP; for now call it GQ.

³ Thus the bulk of this paper centers around surprising coordination of unlikes in comma chains. Of course there are many other well known cases of coordination of unlikes such as *Lee is a linguist and proud of it*, or cases of CP and NP coordination as in *Lee believes that the earth is flat and every other silly thing on the internet*. These, however, are different from the type of coordination of unlikes in this paper; those are all instances where an expression of category X may conjoin with an expression of category Y just in case both are allowed in the same environment. Such cases can be handled with an underspecified category, or a union category (e.g., Morrill 1994, Bayer, 1996). The M&M cases are different: while we get *Barry played the flute* we do not get *Barry played does magic tricks*.

⁴ I am grateful to Dick Oerhle for suggesting this to me and for pointing out that with this there is no mystery as to why Lift preserves word order.

⁵ The view that *and* is polymorphic in the lexicon is quite common, but one might be unhappy at the idea of a lexical entry abbreviating an infinite number of items (since X can range over any category). Jacobson 2014 (Ch. 11) suggests an alternative whereby there is a single item in the lexicon (for S conjunction; possibly a second for N conjunction) and the others are all derived from that by recursive application of a generalized "Geach" rule.

⁶ A common argument given for the asymmetry in conjunction centers on WCO effects (or, binding asymmetries) in conjunction (see, e.g., Munn 1993, Wagner 2005). This assumes that a 'binder' must, in standard terms c-command the pronoun or - in terms of the variable free semantics of, e.g., Jacobson 1999 - occupy an argument slot which is later in the syntactic/semantic composition. Under this view the contrast between *every boy and his mother* vs. **his mother and every boy*, gives evidence (given either of the above ways to think about it) that the first conjunct asymmetrically c-commands (or, is introduced later than) the right conjunct. However, the issues here are complex. First, note that argument rests on the assumption that the phenomenon is not sensitive to a simple left/right asymmetry. But of course pinning this on word order is exactly how this was handled in all the early accounts of WCO type effects (Postal, 1971, Wasow 1972, Jacobson 1972, 1977 ("Leftmost Constraint"), Chomsky 1976 ("Leftness Condition") etc. Therefore, many subsequent authors have again pointed out that the effect could indeed be about word order (Barss and Lasnik 1986, Progovac 1998, Shan and Barker 2006). But the story does not end there, for the linear order account of binding asymmetries is actually not relevant

to the variable free treatment of binding (Jacobson 1999). There the analogue to 'c-command' (i.e., argument order) is not just a stipulation which could as easily be replaced by linear order; it is built into the basic binding apparatus. (An argument slot which is later in the semantic combinatorics is 'merged' with, or 'binds', a slot occupied by a pronoun.) Using word order would be possible but is much less natural; see Jacobson 1999 for discussion.

All that said, we will not take the binding asymmetries to be key evidence for the structure of conjunction, simply because there are there are unclarities as to how to put the semantics together in both the standard and the variable-free accounts. Without a full account, it remains unclear as to what can ultimately be concluded from this asymmetry.

Note too that Progovac 1998 gives additional arguments that the first of the conjuncts does not c-command the second (or - in CG terms - is not introduced as a later in the composition), but these arguments also rest on questionable assumptions about the role of c-command. Her main one centers on Principle C effects; she notes a contrast between (i) and (ii) (the presence and absence of * on these is Progovac's judgments; I find the contrasts much more subtle than good vs. bad):

- (i) *John certainly like's John's wife.
- (ii) John and John's wife are certainly invited.

But these are hardly minimal pairs, and the argument relies on the controversial assumption that this type of "Principle C" violation (also known as the Repeated Name Penalty) is due to the syntax rather than a variety of other factors such as topichood. It is easy enough to construct cases with c-command but where the effect is nonetheless ameliorated:

(iii) ???Lee is invited to birthday parties much more often than Lee's sister is.

And she takes the licensing of a Negative Polarity item to require c-command: "Likewise, a negative word in the first conjunct cannot license a negative polarity item any in the second"; her example is **She chased nobody and/or any dog*. From this she draws the conclusion that the first conjunct cannot c-command the second. But it has been known since at least as early as Ladusaw 1979 that the conditions on the licensing of a Negative Polarity item like *any* are not a simple matter of a negative c-commanding the NPI. *any dog* in the above example (with *and*) is not in a downward entailing environment. (The case with *or* is more complex, but in that case so are the facts.)

⁷ We need to be a bit more precise: $\text{Lift}(a)$ does not denote a unique function because it depends what we are lifting over. Hence when needed we will write $\text{Lift}_{\langle a,b \rangle}(a)$ to mean the function whose domain is the set of functions in $\langle a,b \rangle$.

⁸ Moreover, if an infixation point (or head) is generally inherited from the function, the lifted category will specify that it is inherited from the argument. This means that categories are more fully articulated; they also encode infixation point (or head) inheritance; see Jacobson, Li, and Robertson in preparation.

⁹ Actually, in all cases there are an infinite number of derivations yielding the same meaning, since the subject and the VP can keep lifting over each other.

¹⁰ Jacobson (2014) provides another way also to derive *Lee tells* (with the appropriate meaning) which does not require *Lee* to lift. This involves defining all the rules recursively, and allows for many simpler derivations, but will not affect any of the points here.

¹¹ It has often been argued RNR is much more general and is not just about coordination - a particularly thorough discussion is in Chaves 2007. For example, we find cases like *The student who loves ___ was assigned to mentor the student who hates ___ OT phonology*. We have nothing to say about these cases here and hope that they can be handled in the same way that *wh*-type extraction parasitic gaps are. See Steedman 1987 for an account of parasitic gaps, and even though I am treating (leftwards) *wh*-type extraction differently from the rightward cases the basic idea could be extended to the right cases.

¹² The reader may wonder about cases like *Lee cooked and Sandy ate scallops on Monday and lobster on Tuesday*. This can be put together by having *cook* and *eat* first lift from TVs to category $VP_R(VP_LTV)$. (Recall that TV is an abbreviation for VP_RNP .) As seen earlier, *Lee* and *Sandy* can both lift to the ordinary category for generalized quantifier subjects: S_RVP . Since those can compose with *cook* (and *eat*) by right composition we get *Lee cooked* of category $S_R(VP_LTV)$, and similarly for *Sandy ate*. These two can cojoin, and *Lee cooked and Sandy ate* is of course also of category $S_R(VP_LTV)$. Note that VP_LTV is the category of both a lifted object and also of a 'fancy object' like *scallops in Monday*, as well as being the category of the conjoined fancy object *scallops on Monday and lobster on Tuesday*. It is therefore the right sort of thing to be argument of *Lee cooked and Sandy ate*. The semantics comes out as expected as an interested reader can verify.

¹³ The claim that *scallops on Mondays* or *Lee tells* in *Lee tells lies* can be a constituent often produces a reaction of stunned incredulity: did a proponent of this view not take Introductory syntax where we learn, for example, that *tells lies* is a constituent in *Lee tells lies*? And, after all, these pass none of the standard 'constituent tests'. But this reaction is

unwarranted. In the first place, what we learn is that *tells lies* can be constituent, but the assumption that is the only structure would follow only under a view that an unambiguous sentence has only one structure. Not only is there no reason to assume that, but it would be very difficult to maintain in view of sentences like *Roses are red and violets are blue and irises are white* which almost certainly has two parses. As to the 'tests' - every one of them (except perhaps coordination) diagnoses a necessary but not sufficient condition for constituency; each has additional requirements. Hence note that none of the italicized expressions in (i) pass any of the standard tests except coordination, but are pretty uncontroversially all 'constituents':

- (i) a. I bought every book *which/that had been recommended*.
- b. I think that *the earth is flat*.
- c. I *skated around the rink*.

More thorough discussion can be found in Jacobson 2014, Steedman 2023, Jacobson 2023.

¹⁴ Steedman 2023 does allow for free S-Crossing composition giving a VP/LNP *believes left* but his system precludes this from taking an NP to its left because in his system plain NPs are never introduced into a syntactic derivation: all "NPs" are type raised via a case marker (crucially also there is no free type raising). There is a marker (ACC) which maps an NP to VP/L(VP/RNP) (for object NPs) but there happens to be no case marker yielding a category VP/R(VP/LNP). As Steedman notes, he needs a second ACC marker of category S/L(S/RNP) to account for RNR cases like *Lee loves and Sandy hates Phonology*, somewhat undermining the beauty of having RNR follow immediately from function composition.

¹⁵ We should note that Jacobson 2014 argues for a system without direct function composition but with instead the "Geach" rule (**g**) which is simply a Curried version of function composition. Thus given two functions h (in $B \times C$) and f (in $A \times B$), $\mathbf{g}(h)(f) = h \circ f$. Taking seriously the view of categories here and defining **g** as the Curried version of function composition gives a result analogous to the allowable compositions above. That is (again ignoring infixes) $A/_R B$ can map only into $(A/_R C)/_R (B/_R C)$, and the mirror image is true for $A/_L B$. Since using **g** instead of function composition plays no role here except for the remarks surrounding (43), and since it adds length when expositing the derivations, we will ignore this here and continue to use just function composition. But a reader familiar with the **g** option instead can note that analogous remarks hold in a system adopting **g**.

¹⁶ Pollard's analysis was not embedded within a CG but is easily translated into CG. Because his dissertation is not well known and not readily available, we briefly summarize the key ideas (translating into the system here). Thus take a Dutch case (i):

- (i) ...dat Jan Piet Cecilia zag helpen zwemmen
 ... that Jan Piet Cecilia saw help swim
 "...that Jan saw Piet help Cecilia swim "

The crucial verbs (here *zien* and *helpen*) take an object and a VP or S complement (depending on one's theory); here we follow Pollard and treat the complements as just VPs. As Dutch is SOV, verbs take NP objects to the left. But rather than taking the VP complement either to the left or to the right, the complement is taken as a circumfix, meaning that the control verbs are infixes. Thus (presumably after word order rules apply) a verb like *helpen* is $((S/_L NP)/_L NP)/_C VP$. Pollard takes all expressions to be headed and

states infixation points in terms of heads: in Dutch infixes go to the left of the head (in English - to the right). Moreover, heads are inherited in the expected way. This gives the necessary ingredients; we leave it to the interested reader to have fun working through the example and seeing how this generalizes to the full range of cross-serial dependencies.

¹⁷ Another well-known difference is that RNR does not obey island constraints (McCawley 1982) while of course *wh* extraction does. But since it has often been suggested that the prohibition from extracting out *wh*-constituents is due to processing, it is conceivable that island effects show up when the 'filler' is on the left but not on the right.

¹⁸ Steedman 2023 (fn. 37) claims that this follows from the semantics of *and* and *or*. He adopts the Partee and Rooth semantics (as is also done here). For example, $[[\text{and}]]$ is $\lambda p[\lambda q[p \sqcap q]]$ for p and q variables of any type whose ultimate result is t . He then claims that because $\sqcap$ is only defined for functions of the same type, it follows that p and q (the expected arguments) must be of the same type. This much is correct: $[[\text{and}]]$ expects two arguments of the same type. But why should it not be able to participate in, e.g., function composition just like any other function can? (Or, map to a more complex function by $\mathbf{g}$ as described in fn. 15?). *Before*, for example, also expects two arguments of the same type (in this case propositions) but allows extraction. Thus one instantiation of $(X/X)/X$ is $(VP/VP)/VP$ in which case it denotes a function taking two VP meanings (extensionally, sets) and giving their intersection. Nothing in the semantics stops such a function from composing with a TV type meaning (as is shown by many examples in this paper whose semantics are perfectly coherent). Indeed, if the lack of composition did follow from the semantics (or logical form) there would be no need for a special 'application only' slash.

¹⁹ I thank a referee (hereafter R) for a comment which led me to look at sum *and* in Lakoff chains. R's particular example was slightly different than the one here, and was used to make a different point. Thus R notes the following scenario. Suppose I am in charge of pairing kids in need of adoption with couples, and you ask which kid I will pair with which couple. Let Lee be a kid, and Sandy and Kim a couple. I can answer (with the right prosody) *I'll pair Lee and Sandy and Kim* but not **I'll pair Lee, Sandy, and Kim*. R took this as evidence for a flat structure for comma chains (see Appendix 2) rather than the right branching structure here. Presumably the intuition is that under a flat structure there is no way to collect up two of the disjuncts (here *Sandy and Kim*) to form a couple and then coordinate that (by a pairing interpretation of *and*) with the other disjunct (*Lee*) - giving the desired result. R assumes - as do I - that the version with overt *and*'s can have a right branching structure - so the further assumption here is that with such a structure these can be put together semantically to give the desired reading.

The example is intriguing, but does not lead to the conclusion that comma chains cannot have a right branching structure. For even when all *ands* are overt, the right branching structure yields the correct interpretation only given appropriate semantics for each occurrence of *and*, and any such semantics I can think of blocks a comma chain for independent reasons. Note first that the relevant reading cannot involve sum *and* throughout because sum *and* (like Boolean *and*) is associative; - a string of sum *ands* just gives a group. Perhaps one of the *ands* is sum and the other Boolean. But then this is exactly like the case where the conjunctions are different, and so no comma chain is possible. Notice too that whatever is going on here, the relevant *and* also does not have the

reduced variant 'n, since **I'll pair Lee 'n Sandy 'n Kim* also lacks the relevant reading. (There is sum 'n as we can get chains like *Lee 'n Sandy 'n Kim are a set of triplets.*)

²⁰ There are analyses in which the 'sum' *and* and Boolean *and* are just one item (see, e.g. Champollion 2016 who treats them all as Boolean *and* where the conjuncts have undergone various type shifts for the sum case). If there is a single lexical item here, then of course the above argument to the effect that we would need to be multiplying items to account for the CSC violations does not hold. It is interesting to note that we still need an account of the fact that the interpretation must be uniform throughout. But under any analysis like that of Champollion (where there is one *and* but the types of the conjuncts are different) this will turn out to be a special case of the phenomena discussed in fn. 22.

²¹ There is an interesting interaction between [&] and [v] features and the way in which the categories are instantiated in the lexical specification of *and* and *or*. Assume that *and*, for example, is listed in the lexicon as (X[&]/X)/X for X a variable over categories. The question is: what sorts of categories is X a variable over? In particular, if a full category is a feature bundle, is S[v] an instantiation of the schema above? If so, then all instances of "X" would need the [v] feature. I will claim here that although any instantiation of "X" might include some features, it never includes features like [&] and [v]. To understand why that is the case would require a fuller treatment of features; for now we rely on the intuition that those features are different from, e.g., Case and Agreement features.

The hypothesis that X just stands for categories without any of the features [&] and [v] is motivated by two related cases. First, consider a case like *After college, Lee will go to graduate school, join a fancy company, or [inherit a batch of money and travel the*

world]. (The bracketing is to show the intended reading.). The expression in brackets here has to have the category VP[&]. (Actually that is S[&]/NP but for expository ease I just use VP here.) But if in this case the instantiation of X in the *or* schema - which is (X[v]/X)/X - were VP[&] then we would have a conflict on the category of the underlined material above. The lexical specification of *or* says this should be VP[v], but since the first argument of *or* is VP[&] then both the left argument of *or* should have the [&] feature as should also the 'top' category (the result). Not only does it seem unlikely that those features can co-occur, but we need a way to ensure that only *or* semantics is operative should this combine with another VP to its left in a comma chain. For example, in *Lee will start a teaching career, go to graduate school, join a fancy company, or [inherit a batch of money and travel the world]* the semantics is only the *or* semantics. Were the [&] feature on the expression starting with *go* the grammar should, in theory, allow either semantics.

Fortunately, that result follows from the hypothesis that the instantiation of X in the original *or* schema cannot include one of these features. Here the instance of *or* being used is (VP[v]/VP)/VP. But then how is it that the right argument of *or* can happen to have an [&] feature? This follows from what we have already seen: Anything of category A/B can combine with a B[&] (see discussion on p. 33), and that is exactly what is happening here. That principle allows the lowest VP to happen to have the feature [&] but there is no requirement for matching of that feature either on the left argument or on the result.

The same is true for the left argument. If it happened to have an [&] feature in virtue of containing coordinated material, nothing would cause that feature to be on the result; whatever feature is on the result only comes from the highest *and* or *or* itself. Similarly,

in the example above, the fact that there is [&] on the right disjunct certainly does not force [&] on the left one. That kind of matching is not required because that is not part of what it means to instantiate X in the schema $(X[v]/X)/X$. We should note that Jacobson 2014 suggests that *and* and *or* are not listed in the lexicon as schemas over an infinite number of items, but that there is a single lexical item *and* (for example) of category $(S[&]/S)/S$ with others derived from this by application of a generalized "Geach" rule. The same issues arise there and the same basic solution can be maintained, as an interested reader can verify.

²² There is more to say. A silent ~~conj~~ not only needs the appropriate feature on its right argument, but the types must be consistent throughout the chain. One might hope this follows from the fact that Boolean meet and join are defined only for functions of the same type. Unfortunately, this is not quite enough. The reason is that one can have, e.g., an *and* conjoining lower types at the bottom of the chain, but since the result can lift it ought to be able to conjoin with a higher type further up. To illustrate, consider *every professor and thief*. This can be understood where $[[\text{professor and thief}]]$ refers to a single individual who is in the intersection of $[[\text{professor}]]$ and $[[\text{thief}]]$ (hence Π defined for $\langle e, t \rangle$ functions). But it can also have a reading equivalent to *every professor and every thief*, which is the result of type lifting each of the nouns over determiners (to type $\langle \langle e, t \rangle, e \rangle$) the interested reader can work through the derivations to see that the meaning comes out as claimed. But now consider *every music lover, professor, and thief came to the meeting*. This can be understood as everyone who has all three properties, or as synonymous with *every music lover, every professor, and every thief*, but it has no 'mixed' reading where it is every music lover and everyone who is both a professor and a thief. But this does not

immediately follow from saying that Π is defined only for functions of the same type. The problem is that *professor and thief* can first conjoin giving an $\langle e, t \rangle$ meaning. But that should then be able to lift as above, and then conjoin with the lifted reading of *music lover*.

So the compositional semantics needs to keep track of either the types or the categories, and not just whether the relevant operation is meet or join. One way is to elaborate the feature system, where $[\&]$ and $[v]$ are feature names which can have as values types (or categories). The idea of having categories as feature values is not new - this is, for example, exactly what is done in the GPSG treatment of slash as a feature. In the example above, then, *professor and thief* will have category $N[\&:\langle e, t \rangle]$. Both this and *music lover* can lift over the determiner, but $[[\textit{and}]]$ is defined only as meet of $\langle e, t \rangle$ meanings, not of $\langle et \rangle, \langle \langle et, et \rangle \rangle$ meanings (which is the meaning of Ns lifted over determiner meanings). There are additional questions about how the features should be instantiated under Lift but space precludes a thorough discussion; it is enough to note that keeping track of the type of functions rules out the missing meaning.

A related example (inspired by a referee's comment) is (i):

- (i) *She's a landscaper, happy and tall musician.

Assuming that both adjectives here are of category N/N (see, e.g., Jacobson 2014), this is an instance of the above. The overt *and* conjoins material of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$ (or, category N/N), while the higher *and* conjoins two $\langle e, t \rangle$ meanings (or, two Ns). The feature $[\&:\langle \langle e, t \rangle, \langle e, t \rangle \rangle]$ ensures that *and* takes something of this type, so *happy and tall* is of category $N[\&:\langle et, et \rangle]/N$. When combining with the noun *musician* we will get the category $N[\&:\langle et, et \rangle]$. But $[[\textit{and}]]$ in this case is defined only for meet of types $\langle et, et \rangle$

and won't be able to do anything when it finds [[landscaper]] of type $\langle e, t \rangle$. The point holds equally well for (ii); both conjuncts are of category $N[\&:\langle e, t \rangle]$ so the semantics of *and* requires two meanings of type $\langle e, t \rangle$, so could not conjoin two noun ($\langle e, t \rangle$) meanings.

- (ii) *She's a talented and underpaid landscaper, happy and tall musician.

In view of these remarks a reader might at this point wonder why this paper is full of cases of expressions of some type (e.g., t) conjoining with expressions of a different type (e.g., $\langle e, t \rangle$) by function composition. In fact, every example of a Lakoff chain involves (by definition) cases of two expressions of different types conjoining. But this is no problem because these combine by function composition, not application. As one example, take the composition of the underlined expression in *Barry does magic tricks, plays dulcimer and Patty plays the flute.* *and Patty plays the flute* is of category $S[\&:t]/_L S$. But it composes with an expression of type $\langle e, t \rangle$ rather than taking that as argument, so there is no problem. The semantics of that is $\lambda x[\text{plays-dulcimer}'(x) \sqcap \text{patty-play-flute}']$. This is similar to the remarks in fn. 18.

²³ As is well known, there is a strong principle which has the effect ruling out 'extraction' of a conjunct (rather than just *from* a conjunct). This does not follow immediately from many accounts of extraction including the one here, so we leave this unsolved.

²⁴ One can get structures where *and/and* combines first with the left argument by function composition and then combines with the right argument. Unsurprisingly, the generalization discussed in Sec. 5.2 is not changed by this as the interested reader can verify.

²⁵ Larson 1985, among others, claimed that *either* can go no higher than a tensed S :

- (i) a. I said that either the storm would miss us or it would only graze us.

b.*I said either that the storm would miss us or it would only graze us.

This follows here because the [v] feature would stop at S and not be on CP. The empirical generalization, however, is controversial; Hofmeister 2010 conducted a large scale judgment study which found that *either* can in fact go above embedded Ss. I have no account of his findings, but will note that the one example he gives (taken from actual corpus data) is extremely confusing. I think we would need to look more closely at the examples and also see if the results can be replicated before drawing any firm conclusion.

Along these lines, a referee questions whether (iib) is bad, as is predicted here:

(ii) a. Sarah said that she invited Barry, he came, and brought along his girlfriend.

b. Sarah said that she invited Barry, that he came, and brought along his girlfriend.

I actually do find (iib) bad (admittedly not crashingly so); there is a strong desire to change the third conjunct to *and that he brought along his girlfriend*. But clearly more systematic large scale judgment studies are needed.

²⁶ If it does turn out that binding asymmetries reflect c-command and/or the order of arguments, (i) (spoken by a judge) provides additional evidence against the flat analysis:

(i) a. I want to see every defendant_i, his_i lawyer and the court clerk in my chambers.

b. *I want to see his_i lawyer, every defendant_i, and the court clerk in my chambers.

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Linguistics Program

Brown University

pauline_jacobson@brown.edu
